

Project Data Page:

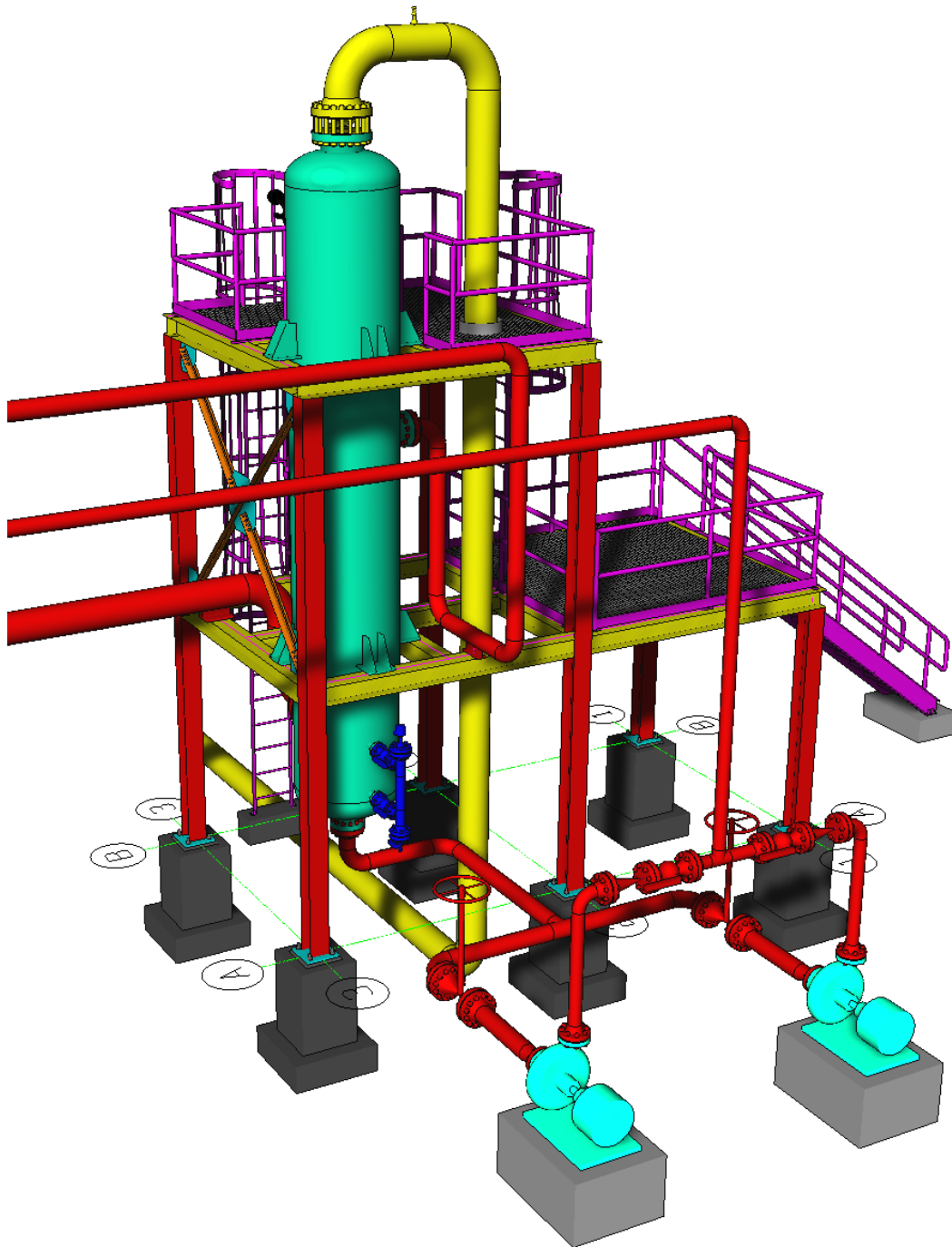




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : D:\ACWA\MR\Cold Exchanger-AK - Copy.pvdb

Date of Analysis : Sep 2, 2026 5:03pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1250.000 mm. ID
 Vessel Design Length, Tangent to Tangent 4960.00 mm.
 Distance of Bottom Tangent above Grade 3000.00 mm.
 Distance of Base above Grade 0.00 mm.
 Specified Datum Line Distance 3155.03 mm.
 Shell Side Design Temperature 85 °C
 Channel Side Design Temperature 85 °C
 Shell Side Design Pressure 191.000 bars
 Channel Side Design Pressure 191.000 bars
 Shop Shell Side Test Pressure 175.386 bars
 Field Shell Side Test Pressure 174.931 bars
 Shop Channel Side Test Pressure 188.489 bars
 Field Channel Side Test Pressure 188.425 bars
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-516 70	...	...	K02700	+IT
Nozzle	SA-266 2	...	...	K03506	
Tubes	SA-210 A-1	...	...	K02707	
Tubesheet	SA-266 2	...	...	K03506	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-193 B8	1	...	S30400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt	...	...	...	1.5000	N/A	No
Bottom Dish	191.037	0.00	194.000	1.5000	N/A	No
Main Shell	191.027	0.00	195.000	1.5000	N/A	No
Top Dish End	191.004	1.03	194.000	1.5000	N/A	No

Liquid Level: 5900.00 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.060

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	-155.0	3000.0	25.0	25.0	...	...	1.00	1.00
Sphere	-155.0	0.0	57.0	48.0	47.2	...	1.00	1.00
Cylinder	4650.0	4650.0	98.0	98.0	96.1	...	1.00	1.00
Sphere	4805.0	0.0	57.0	48.0	47.2	...	1.00	1.00

External Pressure Calculations:



From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	Tsht	48	No Calc	...	81.7573
Tsht	30	48	No Calc	...	81.7573
30	Tsht	98	No Calc	...	108.717
Tsht	40	98	No Calc	...	No Calc
40	50	48	3.50807	1.03419	81.7573

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	3000	No Calc	No Calc	No Calc
20	Tsht	No Calc	No Calc	No Calc	No Calc
Tsht	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	4649.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1500	1697.84	4300.45	7368.86	23995.5
20	30	2999.98	1123.83	4205.72	3136.29	11236.2
30	40	5480	1123.83	3831.24	2962.06	10642.2
40	50	7959.98	125.555	623.756	37.7298	187.441

Abs Max of the all of the Stress Ratio's : 0.9407

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	32.0000	mm.
Inside Diameter of Basing	1246.0000	mm.
Outside Diameter of Basing	1552.0000	mm.
Nominal Diameter of Bolts	27.0000	mm.
Diameter of Bolt Circle	1462.0000	mm.
Number of Bolts	12	
Thickness of Gusset Plates	10.0000	mm.
Average Width of Gusset Plates	96.0000	mm.
Height of Gussets	117.0000	mm.
Distance between Gussets	78.0000	mm.
Thickness of Top Plate or Ring	24.0000	mm.
Radial Width of the Top Plate	117.0000	mm.

Loads for Foundation/Support Design:

Total Wind Shear on Support	1698.	Kgf
Total Earthquake Shear on Support	4300.	Kgf
Wind Moment on Support	7369.	Kg-m.
Earthquake Moment on Support	23996.	Kg-m.
Maximum Initial Bolt Load	2316.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	39192.3 kg.
Shop Test - Fabricated + water (full)	45338.1 kg.
Shipping - Fab. + rem. intls.+ shipping app.	39206.1 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	39206.1 kg.
Empty - Fab. + intls. + details + wghts.	39206.1 kg.
Operating - Empty + operating liquid (no ca)	39574.7 kg.
Field Test - Empty weight + test medium (full)	45351.9 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
B1-Tube Outlet	194	...	...	OK	38.76	OK	Passed
B2-Shell outlet	195	...	...	OK	18.02	OK	Passed
B3-Shell outlet	195	...	...	OK	18.02	OK	Passed
A2-Shell inlet	195	...	...	OK	18.02	OK	Passed
A3-Shell Inlet	195	...	...	OK	18.02	OK	Passed
A1-Tube inlet	192	OK	...	OK	38.75	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 195 Nozzle : B2-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 192 Nozzle : A1-Tube inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
20	B1-Tube Outlet	0.000	0.000	987.151
30	B2-Shell outlet	4055.000	180.000	466.050
30	B3-Shell outlet	4055.000	360.000	466.050
30	A2-Shell inlet	855.000	360.000	466.050
30	A3-Shell Inlet	855.000	180.000	466.050
40	A1-Tube inlet	0.000	0.000	987.151

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
B2-Shell outlet	250 mm	160	None	273.0	28.575	...	...	343	...
B3-Shell outlet	250 mm	160	None	273.0	28.575	...	...	343	...
A2-Shell inlet	250 mm	160	None	273.0	28.575	...	...	343	...
A3-Shell Inlet	250 mm	160	None	273.0	28.575	...	...	343	...
B1-Tube Outlet	600 mm	160	None	609.6	59.512	...	...	389	...
A1-Tube inlet	600 mm	160	None	609.6	59.512	...	...	389	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
B2-Shell ou	SA-266 2	98.000	10.000	...	...	...
B3-Shell ou	SA-266 2	98.000	10.000	...	...	...
A2-Shell in	SA-266 2	98.000	10.000	...	...	...
A3-Shell In	SA-266 2	98.000	10.000	...	...	...
B1-Tube Out	SA-266 2	46.000	10.000	...	...	...
A1-Tube inl	SA-266 2	46.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
B2-Shell outlet	3899.975	180.0	200.00	0.00	Main Shell
B3-Shell outlet	3899.975	360.0	200.00	0.00	Main Shell
A2-Shell inlet	699.975	360.0	200.00	0.00	Main Shell
A3-Shell Inlet	699.975	180.0	200.00	0.00	Main Shell
B1-Tube Outlet	...	0.0	200.00	0.00	Bottom Dish
A1-Tube inlet	...	0.0	200.00	0.00	Top Dish End

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 57.0mm. THK X 1300.0mm. ID X 0.0mm.	SA-516 70
2	INSULATION: 625mm X 10mm THK	
2	TUBESHEET: 1446.0mm X 155.0mm THK	SA-266 2
2893	TUBES: 4960.0mm X 12.7mm DIA X 1.6mm THK	SA-210 A-1
1	BASERING: 1552.0mm. OD X 1246.0 ID X 32.0mm.	SA-516 70
12	BASERING BOLTS: 27.0mm. DIA	SA-193 B8
12	BASERING BOLT NUTS: 27.0mm. DIA	...
1	BASERING TOP RING: 1594.0mm. OD X 1360.0 ID X 24.0mm.	...
24	BASERING GUSSET PLATES: 96.0mm. X 96.0mm. X 117.0mm. X 10.0mm.	...
1	CON SKIRT SEGMENT 25.0mm. THK X 3194.3mm. X 3000.0mm. X 3048.0mm	SA-516 70
1	CON SKIRT SEGMENT 25.0mm. THK X 921.2mm. X 3000.0mm. X 879.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 98.0mm. THK X 3048.0mm. X 4650.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 98.0mm. THK X 879.0mm. X 4650.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Main Shell [8]		B	31	13		0.674	98.000	1.00	Yes
B2-Shell outle[1]		B	31	30		0.980	98.000	1.00	Yes
B3-Shell outle[1]		B	31	30		0.980	98.000	1.00	Yes
A2-Shell inlet[1]		B	31	30		0.980	98.000	1.00	Yes
A3-Shell Inlet[1]		B	31	30		0.980	98.000	1.00	Yes
Tubesheet: SS[13]		B	31	31		1.000	98.000	1.00	Yes
Warmest MDMT:			31	31					
Bottom Dish [9]		B	16	2		0.744	48.000	1.00	Yes
Top Dish End [9]		B	16	2		0.744	48.000	1.00	Yes
B1-Tube Outlet[1]		B	16	15		0.983	48.000	1.00	Yes
A1-Tube inlet [1]		B	16	15		0.983	48.000	1.00	Yes
Tubesheet: CS[14]		B	20	20		1.000	57.000	1.00	Yes
Warmest MDMT:			20	20					
Exchanger Side		Computed MDMT			Required MDMT		Pass/Fail		
Shell		31.0			-20.0		Fail		
Channel/Tube		20.0			-20.0		Fail		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.



Class From To : Basic Element Checks.

Warn. 10 20 Skirt diameter at base is < at the top

Class From To: Check of Additional Element Data

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	191	bars
Channel Side Design Pressure	191	bars
Shell Side Design Temperature	85.0	°C
Channel Side Design Temperature	85.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Lethal	
Service Type, Channel Side	Lethal	
MDMT (CET), Shell Side	-20.0	°C
MDMT (CET), Tube Side	-20.0	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes



Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	0 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
--	-----



Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/4
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt
Distance "FROM" to "TO"	3000 mm.
Skirt Outside Diameter	1360 mm.
Diameter of Skirt at Base	1300 mm.
Skirt Thickness	25 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	25 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	85 °C
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	242 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No

Element#	2/4
Element From Node	20
Element To Node	30
Element Type	Spherical
Description	Bottom Dish
Distance "FROM" to "TO"	0 mm.
Inside Diameter	1300 mm.
Element Thickness	48 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	57 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
-------------------	----



Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	-625	mm.
Height/Length of Liquid	625	mm.
Liquid Density	0.59900E-04	kg./cm ³
Element From Node	20	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	-625	mm.
Height/Length of Insulation	625	mm.
Thickness of Insulation	10	mm.
Density	0.0001918	kg./cm ³
Element From Node	20	
Detail Type	Nozzle	
Detail ID	B1-Tube Outlet	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	600	mm
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2424.4	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-266 2	

Element#	3/4	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	4650	mm.
Inside Diameter	1250	mm.
Element Thickness	98	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	98	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	191	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	4650	mm.
Liquid Density	0.59904E-04	kg./cm ³
Element From Node	30	
Detail Type	Nozzle	
Detail ID	B2-Shell outlet	
Dist. from "FROM" Node / Offset dist	3900	mm.
Nozzle Diameter	250	mm
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	353.05	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-266 2	



Element From Node	30
Detail Type	Nozzle
Detail ID	B3-Shell outlet
Dist. from "FROM" Node / Offset dist	3900 mm.
Nozzle Diameter	250 mm
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	360.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	353.05 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-266 2

Element From Node	30
Detail Type	Nozzle
Detail ID	A2-Shell inlet
Dist. from "FROM" Node / Offset dist	700 mm.
Nozzle Diameter	250 mm
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	360.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	353.05 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-266 2

Element From Node	30
Detail Type	Nozzle
Detail ID	A3-Shell Inlet
Dist. from "FROM" Node / Offset dist	700 mm.
Nozzle Diameter	250 mm
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	353.05 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-266 2

Element From Node	30
Detail Type	Weight
Detail ID	Datasheet diffe
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	1885 Kg
Offset from Element Centerline	0 mm.

Element#	4/4
Element From Node	40
Element To Node	50
Element Type	Spherical
Description	Top Dish End
Distance "FROM" to "TO"	0 mm.
Inside Diameter	1300 mm.
Element Thickness	48 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	57 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
-------------------	----



Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	625 mm.
Liquid Density	0.59900E-04 kg./cm ³
Element From Node	40
Detail Type	Insulation
Detail ID	Hot insulation
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	625 mm.
Thickness of Insulation	10 mm.
Density	0.0001918 kg./cm ³
Element From Node	40
Detail Type	Nozzle
Detail ID	A1-Tube inlet
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	600 mm
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	461.57 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-266 2



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
	Skirt	...	3000	...	3000
	Bottom Dish	...	3000	...	...
	Main Shell	...	7805	...	4650
	Top Dish End	...	7960	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Skirt	...		25	1.5	1360	...	1
Bottom Dish	191.04		57	1.5	1300	138	2
Main Shell	191.03		98	1.5	1250	138	3
Top Dish End	191		57	1.5	1300	138	4

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Skirt	...		No Calc	No Calc	25	No Calc	1
Bottom Dish	191		194.168	200.834	48	47.2309	2
Main Shell	191		194.537	197.762	98	96.0868	3
Top Dish End	191		194.201	200.834	48	47.2229	4

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

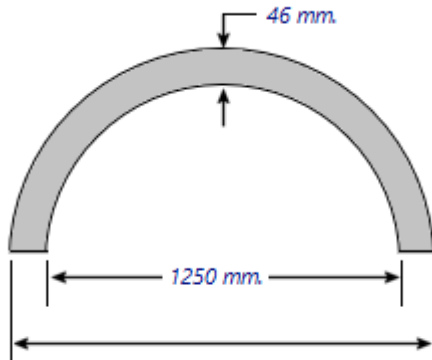
Shell Side MAWP = 135 bars, < design pressure: 191.000 bars
 Shell Side MAPnc = 135 bars
Channel Side MAWP = 145 bars, < design pressure: 191.000 bars
 Channel Side MAPnc = 145 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom Dish

Spherical Head Design Information:					
Design Pressure	P	191.037 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		0.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1300.000 mm.	Actual Stress		135.7 N./mm ²
Spec. Min. Thk t _e , or s _a		48.0000 mm.	Spec. Nominal Thk.		57.000 mm.
Surface Area		31406.4 cm ²	SG of Contents		0.060
Element Volume		575276.3 cm ³	Weight of Contents		34.4 Kgf
Empty Weight		1278.6 Kgf	Operating Weight		1313.0 Kgf
			Comp. All. Ope. Stress		147.3 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 16.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.125 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (191.037 \cdot 651.5) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 191.037)$$

$$= 45.7309 + 1.5000 = 47.2309 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.037 bars

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 46.5) / (651.5 + 0.2 \cdot 46.5)$$

$$= 194 - 0 = 194 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 48.0) / (650.0 + 0.2 \cdot 48.0)$$

$$= 201 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$



$$= (191.037 \cdot (651.5 + 0.2 \cdot 46.5)) / (2 \cdot 1.0 \cdot 46.5)$$
$$= 135.747 \text{ N./mm}^2$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.301 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 48.0$, $t_r = 34.591$, $c = 1.5 \text{ mm.}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.744$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 16 °C

Min Metal Temp. at Required thickness (UCS 66.1) 2 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	144.991 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-20 °C
Governing Thk.	48.000 mm.	Basic MDMT	16 °C
Stress/Thk. Ratio	0.744	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	2 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

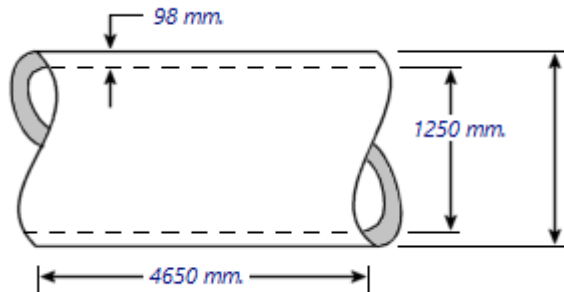
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 16.250 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.125 mm.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	191.027 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		4650.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1250.000 mm.	Actual Stress		135.5 N./mm ²
Spec. Min. Thk t _e , or s _a		98.0000 mm.	Spec. Nominal Thk.		98.000 mm.
Surface Area		211237.6 cm ²	SG of Contents		0.060
Element Volume		4003012.5 cm ³	Weight of Contents		240.0 Kg
Empty Weight		14957.1 Kg	Operating Weight		15197.2 Kg
			Comp. All. Ope. Stress		147.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (191.027 \cdot 626.5) / (138.0 \cdot 1.0 - 0.6 \cdot 191.027)$$

$$= 94.5868 + 1.5000 = 96.0868 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.027 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 96.5) / (626.5 + 0.6 \cdot 96.5)$$

$$= 195 - 0 = 195 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 98.0) / (625.0 + 0.6 \cdot 98.0)$$

$$= 198 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (191.027 \cdot (626.5 + 0.6 \cdot 96.5)) / (1.0 \cdot 96.5)$$

$$= 135.489 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 7.270 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 98.0, tr = 65.07, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.674$, Temp. Reduction = 18 °C

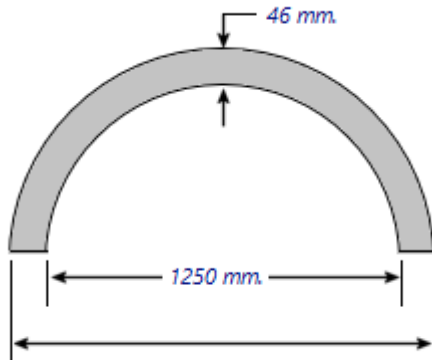
Min Metal Temp. w/o impact per UCS-66, Curve B 31 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 13 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was
 specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	134.913 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-20 °C
Governing Thk.	98.000 mm.	Basic MDMT	31 °C
Stress/Thk. Ratio	0.674	Temperature Difference	18 °C
UCS-66(c) red.	No	Computed Min. Temp.	13 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top Dish End

Spherical Head Design Information:					
Design Pressure	P	191.004 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		0.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1300.000 mm.	Actual Stress		135.7 N./mm ²
Spec. Min. Thk t _e , or s _a		48.0000 mm.	Spec. Nominal Thk.		57.000 mm.
Surface Area		31406.4 cm ²	SG of Contents		0.060
Element Volume		575276.3 cm ³	Weight of Contents		34.4 Kg
Empty Weight		1278.6 Kg	Operating Weight		1313.0 Kg
			Comp. All. Ope. Stress		147.3 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 16.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.125 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (191.004 \cdot 651.5) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 191.004)$$

$$= 45.7229 + 1.5000 = 47.2229 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.004 bars

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 46.5) / (651.5 + 0.2 \cdot 46.5)$$

$$= 194 - 0 = 194 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 48.0) / (650.0 + 0.2 \cdot 48.0)$$

$$= 201 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (191.004 \cdot (651.5 + 0.2 \cdot 46.5)) / (2 \cdot 1.0 \cdot 46.5)$$

$$= 135.723 \text{ N./mm}^2$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.301 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 48.0$, $t_r = 34.591$, $c = 1.5 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.744$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 16 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 2 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	144.991 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-20 °C
Governing Thk.	48.000 mm.	Basic MDMT	16 °C
Stress/Thk. Ratio	0.744	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	2 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 16.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.125 mm.

EXTERNAL PRESSURE RESULTS for: Top Dish End

Spherical Head From 40 to 50 Ext. Chart: CS-2 at 85 °C

Elastic Modulus from Chart: CS-2 at 85 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Sphere OD	Do/t	Factor A	Factor B
46.500	1396.00	30.02	0.0083274	122.73

$$MAEP = B / ((Do/t)/2) = 122.731 / (30.0215/2) = 81.7573 \text{ bars}$$

Results for Required Thickness				Tca
Tca	Sphere OD	Do/t	Factor A	Factor B
2.008	1396.00	695.20	0.0003596	35.95

$$MAEP = B / ((Do/t)/2) = 35.953 / (695.1953/2) = 1.0343 \text{ bars}$$

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	175.386 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	175.259 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	148.404 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	134.913 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	188.489 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	188.361 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	159.490 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	144.991 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish	129.6	234.0	0.554	188.62
Main Shell	122.5	234.0	0.523	175.51
Top Dish End	129.6	234.0	0.554	188.62

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B1-Tube Outlet	Nozzle	138.00	138.00	1.000
B2-Shell outlet	Nozzle	138.00	138.00	1.000
B3-Shell outlet	Nozzle	138.00	138.00	1.000
A2-Shell inlet	Nozzle	138.00	138.00	1.000
A3-Shell Inlet	Nozzle	138.00	138.00	1.000
A1-Tube inlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Dish	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Top Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	118.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
B1-Tube Outlet	89.06	225.00	0.396
B2-Shell outlet	76.84	225.00	0.342
B3-Shell outlet	76.84	225.00	0.342
A2-Shell inlet	76.84	225.00	0.342
A3-Shell Inlet	76.84	225.00	0.342
A1-Tube inlet	89.06	225.00	0.396

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	175.386 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	174.931 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	148.404 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	134.913 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	188.489 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	188.425 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	159.490 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	144.991 bars

Vertical Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish	134.0	234.0	0.573	188.58
Main Shell	124.4	234.0	0.532	175.39
Top Dish End	133.9	234.0	0.572	188.49

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B1-Tube Outlet	Nozzle	138.00	138.00	1.000
B2-Shell outlet	Nozzle	138.00	138.00	1.000
B3-Shell outlet	Nozzle	138.00	138.00	1.000
A2-Shell inlet	Nozzle	138.00	138.00	1.000
A3-Shell Inlet	Nozzle	138.00	138.00	1.000
A1-Tube inlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Dish	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Top Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	118.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
B1-Tube Outlet	89.05	225.00	0.396
B2-Shell outlet	76.78	225.00	0.341
B3-Shell outlet	76.78	225.00	0.341
A2-Shell inlet	76.78	225.00	0.341
A3-Shell Inlet	76.78	225.00	0.341
A1-Tube inlet	89.00	225.00	0.396

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		174.93 bars
Shell Thickness	ts	98.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1250.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Bottom Dish

Channel Type:		Hemispherical Head
Channel Test Pressure		188.42 bars
Channel Thickness	tc	57.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1300.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	2893
Tube Wall Thickness	et	1.6500 mm.
Tube Outside Diameter	D	12.7000 mm.
Total Straight Tube Length	Lt	4960.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		4650.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-210 A-1
Tube Material UNS Number		K02707
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	255.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	19.0500 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm.
Groove Weld Leg	ag	1.6500 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	536.453 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.
Area of Tube Layout	Ap	8856.5 cm ²
Perimeter of Tube Layout	Cp	3478.0 mm.

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 155.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm.
 Tubesheet Outside Diameter A 1446.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

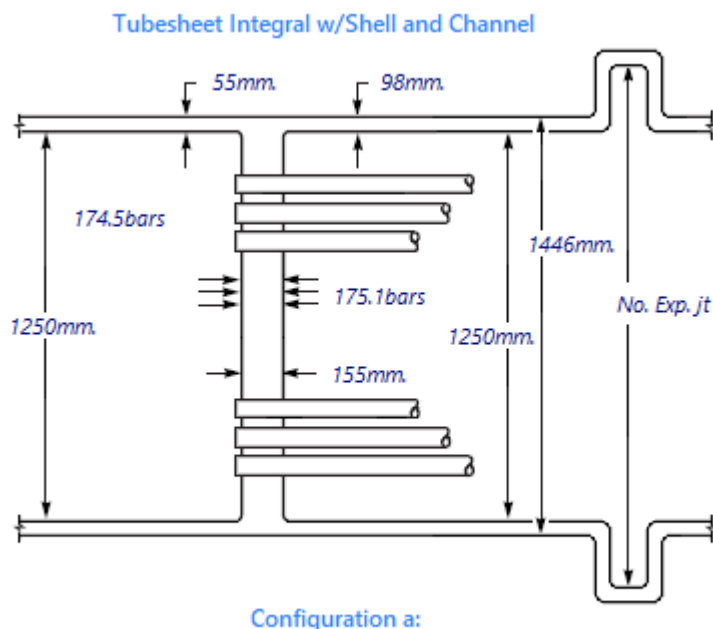
Unsupported Tube Span under consideration l 500.000 mm.
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design Yes
 Differential Design Pressure Pdif 20.0000 bars
 Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
 Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len 21.1 °C 0.0000115191 /°C



Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Differential pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (20.0 \cdot (12.7/2 - 0)) / (118.0 \cdot 1.0 + 0.4 \cdot 20.0)$
 $= 0.1069 + 0.0000 = 0.1069 \text{ mm.}$

Tube Required Thickness under External Pressure (Differential pressure) :

External Pressure Chart CS-2 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	12.70	4650.00	7.70	50.0000	0.0185675	122.73

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 243.2871 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3034	12.70	4650.00	41.85	50.0000	0.0006280	62.78

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 62.7848) / (3 \cdot 41.8524) = 20.0008 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3034 mm.
Allowable Internal Pressure at Corroded thickness	342.15 bars
Required Internal Design Pressure	20.00 bars
Allowable External Pressure at Corroded thickness	243.29 bars
Required External Design Pressure	20.00 bars
Required Thickness due to Shell Side pressure	0.3034 mm.

Note: The tubes were evaluated using the specified differential pressure.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Note: Shell/Channel Design Pressures modified by Input differential pressure:
 Psd = 20.000 Ps = 168.425 Pt = 188.425 bars

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/155.0$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d^*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(12.7 - 2 \cdot 1.65 (0.20270E+09/2.0270E+09) \cdot (117/137) \cdot (0); 12.7 - 2 \cdot 1.65)$$

$$= 12.7000 \text{ mm.}$$

Equivalent outer tube limit circle diameter [D_o]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 536.453 + 12.7$$

$$= 1085.606 \text{ mm.}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (19.05 - 12.7) / 19.05$$

$$= 0.333$$

Equivalent outer tube limit radius [a_o]:

$$= D_o / 2$$

$$= 1085.606/2$$

$$= 542.8030 \text{ mm.}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 19.05 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 1085.606 \cdot 19.05) / (\pi \cdot 1085.606^2)) }$$

$$= 19.0500 \text{ mm.}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (19.05 - 12.7) / 19.05$$

$$= 0.3333$$

Ratio [ρ_s]:

$$= a_s / a_o$$

$$= 625.0/542.803$$

$$= 1.1514$$

Ratio [ρ_c]:

$$= a_c / a_o$$

$$= 650.0/542.803$$

$$= 1.1975$$

Compute parameter [x_t]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 2893 ((12.7 - 2 \cdot 1.65) / (2 \cdot 542.803))^2$$

$$= 0.7831$$

Determine parameter [x_s]:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 2893 (12.7 / (2 \cdot 542.803))^2$$

$$= 0.6041$$

Determine the value [$h'g$]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((6.0 - 0); 0)$$

$$= 6.000 \text{ mm.}$$



ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts (Ds + ts) Es / L$$

$$= \pi \cdot 98.0 (1250.0 + 98.0) \cdot 2.0270E+09 / 4650.0$$

$$= 1844870.3750 \text{ Kg/mm.}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt (Dt - tt) Et / L$$

$$= \pi \cdot 1.65 (12.7 - 1.65) \cdot 2.0270E+09 / 4650.0$$

$$= 254.6221 \text{ Kg/mm.}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 1844870 / (2893 \cdot 254.622)$$

$$= 2.50450$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kt)$$

$$= 1 / (1 + 1844870/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1250.0 + 98.0)98.0)$$

$$= 0.0050 \text{ 1/mm.}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.005 \cdot 0.20270E+09 \cdot 98.0^3 / (6(1 - 0.3^2))$$

$$= 1747590144.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1250.0 + 98.0) \cdot 0.17476E+1 / 155.0^3 \cdot (1 + 155.0 \cdot 0.005 + (155.0 \cdot 0.005)^2 / 2)$$

$$= 7878687.5000 \text{ bars}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 1250.0^2 / (4 \cdot 0.20270E+09 \cdot 98.0) \cdot (1 - 1250.0 \cdot 0.3 / (2(1250.0 + 98.0)))$$

$$= 0.16928E-01 \text{ mm./N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((Dc + tc)tc)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1300.0 + 57.0)57.0)$$

$$= 0.0065 \text{ 1/mm.}$$

Channel coefficient [kc]:

$$= \beta_c \cdot Ec \cdot tc^3 / (6(1 - \nu_c^2))$$

$$= 0.007 \cdot 0.20270E+09 \cdot 57.0^3 / (6(1 - 0.3^2))$$

$$= 449383008.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1300.0 + 57.0 / \cos(0)) \cdot 0.44938E+09 / 155.0^3 \cdot (1.0 + 155.0 \cdot 0.007 \cdot (1.0 + 1.0) / 2 + (155.0 \cdot 0.007)^2 / 2 \cdot 1.0)$$

$$= 2482225.8 \text{ bars}$$

Channel coefficient [δ_c]:

$$= Dc^2 / (4 \cdot Ec \cdot Tc) \cdot (1/2 - Dc \cdot \nu_c / (2(Dc + tc)))$$

$$= 1300.0^2 / (4 \cdot 0.20270E+09 \cdot 57.0) \cdot (1/2 - 1300.0 \cdot 0.3 / (2(1300.0 + 57.0)))$$

$$= 0.13028E-01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$



ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 8.136483$; $\mu^* = 0.333333$; $E^* = 65896428$ KPa.
 $E^*/E = 0.325091$; $\nu^* = 0.334167$

Note: As h/p (8.136) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:
 $= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$
 $= ((24(1 - 0.334^2)2893 \cdot 0.20270E+09 \cdot 1.65$
 $(12.7 - 1.65)542.803^2) / (65896428 \cdot 4650.0 \cdot 155.0^3))^{1/4}$
 $= 2.7698$

Values from Table 4.18.3

$Z_d = 0.068773$; $Z_v = 0.120115$; $Z_m = 0.546768$
 $Z_a = 0.166668E+01$; $Z_w = 0.120115$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio $[K]$:

$= A / D_o$
 $= 1446.0/1085.606$
 $= 1.3320$

Compute coefficient $[F]$:

$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$
 $= (1 - 0.33) / 65896428 \cdot (7878687 + 2482225 +$
 $0.20270E+09 \cdot \ln(1.33))$
 $= 11.0560$

Compute parameter $[\phi]$:

$= (1 + \nu^*) F$
 $= (1 + 0.3342) 11.056$
 $= 14.7506$

Compute parameter $[Q_1]$:

$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$
 $= (1.1514 - 1 - 14.7506 \cdot 0.1201) / (1 + 14.7506 \cdot 0.5468)$
 $= -0.178744$

Compute Parameter $[Q_{z1}]$:

$= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4$
 $= (0.06877 + -0.17874 \cdot 0.12012) / 2 \cdot 2.76983^4$
 $= 1.3921$

Compute parameter $[Q_{z2}]$:

$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4$
 $= (0.12012 + -0.17874 \cdot 0.54677) / 2 \cdot 2.76983^4$
 $= 0.6587$

Compute parameter $[U]$:

$= (Z_w + (p_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$
 $= (0.1201 + (1.1514 - 1) 0.5468) 2.76983^4 / (1 + 14.7506 \cdot 0.5468)$
 $= 1.3175$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor $[\gamma_b]$:

$= 0$ (config a)

Compute Parameter $[\gamma]$:

$= 0.000$ mm. (For Pressure only cases)

Calculate parameter $[\omega_s]$:

$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$
 $= 1.1514 \cdot 0.17476E+1 \cdot 0.005 \cdot 0.016928 (1 + 155.0 \cdot 0.005)$
 $= 30245.5820$ mm.²

Calculate parameter $[\omega^*s]$:

$= ao^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s$



$$= 542.803^2 (1.151^2 - 1) \cdot (1.151 - 1) / 4 - 30245.582$$

$$= -26611.6309 \text{ mm.}^2$$

Parameter [wc]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.197 \cdot 0.44938E+09 [0.013028 (1.0 + 0.007 \cdot 1.0 \cdot 155.0) -$$

$$0 (1.0/155.0 + 1.0 \cdot 0.007/2)]$$

$$= 9225.6572 \text{ mm.}^2$$

Parameter [ω*c]:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega c$$

$$= 542.80298^2 [(1.19749^2 + 1) \cdot (1.19749 - 1) / 4 - (1.15143 - 1) / 2] - 9225.65723$$

$$= 3872.3191 \text{ mm.}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s -$$

$$[(\rho_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 168.425 \{ 0.604 + 2(1 - 0.604) 0.3 + [2/2.505 \cdot$$

$$(1250.0/1085.606)^2] 0.3 - [(1.151^2 - 1)/(1.0 \cdot 2.505)] -$$

$$[(1 - 1.0)/(2 \cdot 1.0 \cdot 2.505)] [(0^2 - (1250.0)^2)/1085.606^2] \}$$

$$= 173.3372 \text{ bars}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.783 + 2(1 - 0.783) 0.3 + 1/(1. \cdot 2.505)] 188.425$$

$$= 247.3116 \text{ bars}$$

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 2893 \cdot 254.622 \cdot 0 / (3.142 \cdot 542.803^2)$$

$$= 0.0000 \text{ bars}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 1.317 \cdot 0 / (2 \cdot \pi \cdot 542.803^2)$$

$$= -0.0000 \text{ bars}$$

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t)$$

$$= -(1.317/542.803^2) \cdot (-26611.631 \cdot 168.425 - 3872.319 \cdot 188.425)$$

$$= 23.3046 \text{ bars}$$

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega^* s \cdot P^* s - \omega^* c \cdot P^* c)$$

$$= (1.317/542.803^2) \cdot (30245.582 \cdot 0 - 9225.6572 \cdot 0)$$

$$= 0.0000 \text{ bars}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (\rho_s - 1) Qz2)) \cdot$$

$$(P^* s - P^* t + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 2.505 / (1 + 1.0 \cdot 2.505 (1.392 + (1.151 -$$

$$1) 0.659)) \cdot (173.337 - 247.312 + 0 + 0 + 0 + 23.305)$$

$$= -26.7934 \text{ bars}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^* s \cdot P_s - \omega^* c \cdot P_t) - (\omega^* s \cdot P^* s - \omega^* c \cdot P^* c)) +$$

$$W^* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-26611.631 \cdot 168.425 - 3872.319 \cdot 188.425) -$$

$$(30245.582 \cdot 0 - 9225.657 \cdot 0)) \cdot 0.01 +$$



$$= 0 \cdot 0 / (2 \cdot \pi) / (1 + 14.7506 \cdot 0.54677) \\ = -5862.7388 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\ = -0.179 + 2 \cdot -5862.739 / (-26.793 \cdot 542.803^2) \\ = -0.03309$$

Fm Value from Table 4.18.3 = 0.096258

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe \\ = (1.5 \cdot 0.0963 / 0.3333) \cdot (2 \cdot 542.803 / (155.0 - 6.0))^2 \cdot -26.79 \\ = -61.6135 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 250.0 = 356.24 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ f]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\ = (1.5 \cdot 0.0294 / 0.3333) \cdot (2 \cdot 542.803 / (28.977 - 6.0))^2 \cdot -12.04 \\ = -356.2368 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc \\ = 28.9774 + 0 + 0 \\ = 28.9774 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy) \\ = (2 \cdot 0.333 \cdot 155.0 / 542.803) \cdot 189.997 \\ = 36.1696 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm.} \\ \text{Tubesheet Shear Stress} = 70.9975 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(HreqB; HreqS) \\ = \max(28.9774; 3.81) \\ = 28.9774 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$Ftmin = -0.0109, Ftmax = 1.907$$

First Extreme Tube Axial Stress from among all the tubes [σ t1]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmin) / (Xt - Xs) \\ = ((168.42 \cdot 0.6041 - 188.42 \cdot 0.7831) - (-26.793) \cdot -0.011) / \\ (0.7831 - 0.6041) \\ = -25.7556 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ t2]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmax) / (Xt - Xs) \\ = ((168.42 \cdot 0.6041 - 188.42 \cdot 0.7831) - (-26.793) \cdot 1.907) / \\ (0.7831 - 0.6041) \\ = 2.9496 \text{ N./mm}^2$$

Maximum Tube Axial Stress [σ t,max]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 25.756 \text{ N./mm}^2$$

Allowable Tube Stress [σ t,A]:

$$= \beta T \cdot Sy,t = 242.2456 \text{ N./mm}^2$$



Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned} &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\ &= \max((3.25 - 0.25(0.069 + -0.033 \cdot 0.12)2.77^4); 1.25) \\ &= 2.0000 \text{ (Should be } \leq 2 \text{)} \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4 \\ &= (\sqrt{ 12.7^2 + (12.7 - 2 \cdot 1.65)^2 }) / 4 \\ &= 3.9501 \text{ mm.} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot E_t / S_yt) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 0.20270E+09 / 254) } \\ &= 129.0317 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 1.0 \cdot 500.0 / 3.95 \\ &= 126.5797 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (S_y / F_s) \cdot (1 - F_t / (2 \cdot C_t)) \\ &= (254 / 2.0) \cdot (1 - 126.58 / (2 \cdot 129.032)) \\ &= 242.246 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)} \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 25.76 \cdot 0.5728 \\ &= 150.43 \text{ Kg} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (12.7 - 1.65) 1.65 \\ &= 0.573 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 1.65 (12.7 - 1.65) 242.246 \\ &= 1414.892 \text{ Kg} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)} \\ &= 0.85 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65) 237.5 \\ &= 1414.8922 \text{ Kg} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 242.246 / 237.496 \\ &= 1.020 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)} \\ &= 0.55 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65) 237.5 \\ &= 953.1818 \text{ Kg} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 (\sqrt{ (0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d } - 0.75 \cdot d_o) \\ &= 2 (\sqrt{ (0.75 \cdot 12.7)^2 + 1.07 \cdot 1.65 \cdot (12.7 - 1.65) 1.02 \cdot 1.0 } - 0.75 \cdot 12.7) \\ &= 1.986 \text{ mm.} \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 0.9928 mm.
 Minimum Required Groove Weld Leg [agr]: 0.9928 mm.

Tube-Tubesheet Jt allowable, 1414.89 is >= tube strength 1414.89 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot S_{y,s} = 0.95 \cdot 260.0 = 370.49 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σ_s ,m]:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 542.803^2 / ((1250.0 + 98.0) 98.0) \cdot [-26.79 + (1.151^2 - 1)$
 $(168.42 - 188.42)] + 625.0^2 \cdot 188.42 / ((1250.0 + 98.0) 98.0)$
 $= 48.2901 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 98.0 mm., the minimum Shell length adjacent to the tubesheet should be 630.000 mm.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	61.6 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	13.2 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	25.8 <=	242.2 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-25.8 <=	-242.2 N./mm ²	Pass
Maximum Force on any one Tube	150.4 <=	1414.9 Kgf	Pass
Axial Membrane Stress in Shell	48.3 <=	370.5 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	28.977	155.000	Ok
Tube-Tubesheet Fillet Weld Leg :	0.993	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	0.993	1.650	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D3uc	28.977 ...	62	356 13 190	Ps+Pt-Th D3	Ok
Max:	28.9774 ... mm.	0.173	0.070 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures



[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D3uc	48	370	...	...	:	...	...	...	...	:		Ok
Max RATIO	0.130		...		:	...		...		:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D3uc	26	242	-26	-242	150	1414	...	370	...	370	Ok
Max RATIO	0.106		0.106		0.106		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	28.977	155.000	Ok
Tube thickness :	0.303	1.650	Ok
Tube-tubesheet fillet weld leg :	0.993	1.650	Ok
Tube-tubesheet groove weld leg :	0.993	1.650	Ok

Min Shell length of thk, (98.000) adj. to tubesheet: 630.000 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm.
 Due to Pressure Shell Compresses by : -0.424 mm.
 Due to Pressure + Thermal Shell Compresses by : -0.424 mm.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0895	...	...	...	...	...
2	-0.4242	...	...	...	...	...
3	-0.3189	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	3000	...	...	No Calc	No Calc
20	Tsht	No Calc	1396	46.5	0.0083274	122.731
Tsht	30	No Calc	1396	46.5	0.0083274	122.731
30	Tsht	4649.97	1446	96.5	0.0066214	122.187
Tsht	40	No Calc	...	96.5	No Calc	No Calc
40	50	No Calc	1396	46.5	0.0083274	122.731

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	Tsht	48	No Calc	...	81.7573
Tsht	30	48	No Calc	...	81.7573
30	Tsht	98	No Calc	...	108.717
Tsht	40	98	No Calc	...	No Calc
40	50	48	3.50807	1.03419	81.7573

Minimum

82

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	3000	No Calc	No Calc	No Calc
20	Tsht	No Calc	No Calc	No Calc	No Calc
Tsht	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	4649.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	2667.83	...	2598.19	...	266.783
20	30	1278.61	575276	1247.68	579268	127.861
30	40	14957.1	4003012	14744.6	4030440	1495.71
40	50	1278.61	575276	1247.68	579268	127.861
Total		20182	5153565.00	19838	5188977.00	2018



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
20	Liqd	34.4177	...	-312.5	...	Liquid: 10
20	Insl	6.883	...	-312.5	...	Hot insulation
20	Nozl	2666.85	...	-726	...	B1-Tube Outlet
30	Liqd	240.034	...	2325	...	Liquid: 20
30	Nozl	388.351	-725	3900	...	B2-Shell outlet
30	Nozl	388.351	725	3900	...	B3-Shell outlet
30	Nozl	388.351	725	700	...	A2-Shell inlet
30	Nozl	388.351	-725	700	...	A3-Shell Inlet
30	Wght	1885	...	...	...	Datasheet diffe
40	Liqd	34.4177	...	312.5	...	Liquid: 30
40	Insl	6.883	...	312.5	...	Hot insulation
40	Nozl	507.729	...	726	...	A1-Tube inlet
20	FTsh	1685.82	...	77.5	...	
20	Tube	7007.27	...	2480	...	Tubes
20	RTsh	1685.82	...	4882.5	...	Tubesheet

Total Weight of Each Detail Type:

Liquid	308.9
Insulation	13.8
Nozzles	4728.0
Weights	1885.0
Exchanger Components	10378.9
Liquid in Tubes	59.7

Sum of the Detail Weights 17374.3 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	22200.4	22200.4	22200.4	22200.4	22200.4	22200.4
Nozzles	4728.0	4728.0	4728.0	4728.0	4728.0	4728.0
Wld Weights	1885.0	1885.0	1885.0	1885.0	1885.0	1885.0
Exchanger	10378.9	10378.9	10378.9	10378.9	10378.9	10378.9
Insulation	...	...	13.8	13.8	13.8	13.8
Ope. Liquid	...	...	...	...	...	308.9
Tube Ope Lqd	...	...	...	...	...	59.7
Test Liquid	...	5150.4	...	...	...	...
Tube Tst Lqd	...	995.4	...	...	...	...
Totals	39192.3	45338.1	39206.1	39206.1	39206.1	39574.7

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	39192.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	45338.1 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	39206.1 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	39206.1 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	39206.1 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	39574.7 kg.
Field Test Wt.	- Empty weight + test medium (full)	45351.9 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	11541.3 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	5436.8 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	11537.5 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	995986.5 cm ³
Weight of Ope Liq in tubes :	59.7 kg.
Weight of Water in tubes :	995.4 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	128177	123465
20	30	31406.4	26546.5
30	40	211238	182605
40	50	31406.4	26546.5

Total	402227.4	359162.7 cm ²
Total	433.0	386.6 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	2934.61	2934.61	2934.61	...	2934.61
20	30	5766.03	5800.45	6340.96	...	5766.03
30	40	26898.5	27198.3	31894.5	...	26898.5
40	50	3606.91	3641.33	4181.83	...	3606.91

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	39206.1	39574.7	45351.9
20	30	36271.5	36640.1	42417.3
30	40	30505.4	30839.6	36076.3
40	50	3606.91	3641.33	4181.83

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment



From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	...	...	...

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle	Flange Rating		Des.			Equiv.	- - - -	Max Pressure	
Description	Ope.	Ambient	Temp	Class	Grade/Group	Press	UG-44	50%	DNV

Shellside Flange Rating

Channelside Flange Rating

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 6.62871 Hz.

The Natural Frequency for the Vessel (Ope...) is 6.59367 Hz.

The Natural Frequency for the Vessel (Filled) is 6.14949 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	0 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.17 \cdot 0.955) / (1 + 1.7 \cdot 3.4 \cdot 0.17)))$
 $= \min(0.85, 0.904)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	6.594 Hz
Natural Frequency of Vessel (Empty)	6.629 Hz
Natural Frequency of Vessel (Test)	6.149 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.792
Structure Height to Diameter ratio	6.533
Height to top of Structure	8668.000 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Zg = 589788$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1511.719 + 0.0 = 1511.719$ mm.
 $= 4.96$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (4.96 ft.) < 15 ft.
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.792 \cdot 51.538$$

$$= 1150.4 \text{ lbs. } [521.8 \text{ Kg}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Skirt	1511.7	0.000	0.000	0.000	1.035	1.000	161.849
Bottom Dish	3000.0	0.000	0.000	0.000	1.035	1.000	161.849
Main Shell	5480.0	0.000	0.000	0.000	1.068	1.000	167.030
Top Dish End	8260.5	0.000	0.000	0.000	1.147	1.000	179.386

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Un corroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kg
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
 Z - Maximum amplitude of vibration at top of vessel, mm.
 DL - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.320E-06 (High); 0.320E-06 < 0.400E-06 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0.0/7960.0 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[4.42/26.12^2] \cdot 10^4 = 64.748$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.400E-06$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 39196 / (7960.0 \cdot 1446.0^2)$$

$$= 0.23550E-05$$

Since Vp is > 0.400E-06 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kg
----	------------------------------	------------------	--------------------------------	-------------

Datasheet dif 0.00 3155.00 162.17 0.00

The Natural Frequency for the Vessel (Ope...) is 6.59367 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1511.72	1596	47880	161.849	574.009
20	30	2999.98	1699.2	0.51792	161.849	0.006209
30	40	5480	1735.2	80686.8	167.03	998.273
40	50	8260.47	1699.2	9449.12	179.386	125.555

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22
Importance Factor		1.500
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(28.2480.75)$
= 0.245 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/6.594)
= 0.152

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.245$, 0.152) per 12.8.2
= 0.152

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= $0.217 / (3.0 / 1.5)$
= 0.109

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= $0.093 / (0.152(3.0 / 1.5))$
= 0.306

Minimum value of C_s per equation 15.4-1 [Cs]:
= $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
= $\max((0.044 \cdot 0.2173 \cdot 1.5), 0.030)$
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= $0.1087 \cdot 39574.7$
= 4300.449 Kgf

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.217 \cdot 39575$

= 1720.2 Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related
to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 6.59367 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	1500	2934.61	94.7293	94.8053	1720.18
20	30	2999.98	5800.45	374.475	372.552	1592.62
30	40	5480	27198.3	3207.49	3174.68	1340.5
40	50	7959.98	3641.33	623.756	618.356	158.276



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1500	1697.84	4300.45	7368.86	23995.5
20	30	2999.98	1123.83	4205.72	3136.29	11236.2
30	40	5480	1123.83	3831.24	2962.06	10642.2
40	50	7959.98	125.555	623.756	37.7298	187.441

Note:
The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	1697.84	0.017133	0.060784	0.35605E-04
20	30	1123.83	0.060785	0.060785	0.35605E-04
30	40	1123.83	0.14665	0.23539	0.38305E-04
40	50	125.555	0.23539	0.23539	0.38305E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	188.913	1180.7
20	30	201.128	1257.05
30	40	205.389	1283.68
40	50	201.128	1257.05

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)

Allowable deflection : 39.800 Actual deflection : 0.235 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	1001.38	942.408	31317284	29540934
20	30	2032.74	1971.41	66231684	64371188
30	40	4150.17	4091.2	131071040	129474048
40	50	2032.74	1971.41	66231684	64371188

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	165.596	280.795	-141.788	-141.788
20	30	165.597	280.795	-147.277	-147.277
30	40	165.597	280.795	-147.277	-147.277
40	50	165.597	280.795	-147.277	-147.277

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	129.99	...	128.238
30	40	58.1842	...	53.289
40	50	129.99	-0.80299	128.238

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-4.07984	-4.1182	-4.71938	...	...
20	30	-1.80434	-1.80731	-1.80434	...	...
30	40	-0.73123	-0.73266	-0.73123	...	...
40	50	-0.17943	-0.17943	-0.17943	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	2.44622	7.96574	0.80725	...
20	30	...	0.4778	1.71177	0.15767	...
30	40	...	0.22435	0.80606	0.074036	...
40	50	...	0.005748	0.028556	0.0018968	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	7.88938
20	30	...	...	...	1.69505
30	40	...	...	...	0.7982
40	50	...	...	...	0.028309

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	0.179
20	30	...	0.079226
30	40	...	0.032132
40	50	...	0.0078735

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-6.53	141.79		0.0460
20		165.60	-2.28	147.28		0.0155
30		165.60	-0.96	147.28		0.0065
40		165.60	-0.19	147.28		0.0013

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	3.99	165.60	-11.79	141.79	0.0241	0.0832
20		165.60	-3.42	147.28		0.0232
30	0.10	165.60	-1.50	147.28	0.0006	0.0102
40		165.60	-0.20	147.28		0.0014

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-6.56	141.79		0.0463
20		165.60	-2.29	147.28		0.0155
30		165.60	-0.96	147.28		0.0065
40		165.60	-0.19	147.28		0.0013

Analysis of Load Case 4 : NP+OW+EQ+FS+BS



From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.03	165.60	-11.90	141.79	0.0243	0.0840
20		165.60	-3.44	147.28		0.0234
30	0.11	165.60	-1.51	147.28	0.0006	0.0102
40		165.60	-0.20	147.28		0.0014

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		280.79	-5.53	141.79		0.0390
20		280.79	-1.96	147.28		0.0133
30		280.79	-0.81	147.28		0.0055
40		280.79	-0.18	147.28		0.0012

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		280.79	-4.72	141.79		0.0333
20		280.79	-1.80	147.28		0.0123
30		280.79	-0.73	147.28		0.0050
40		280.79	-0.18	147.28		0.0012

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-6.56	141.08		0.0465
20	128.66	165.60		147.28	0.7770	
30	57.68	165.60		147.28	0.3483	
40	129.82	165.60		147.28	0.7839	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.03	165.60	-11.90	141.08	0.0243	0.0844
20	129.97	165.60		147.28	0.7849	
30	58.29	165.60		147.28	0.3520	
40	129.85	165.60		147.28	0.7841	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-6.56	141.08		0.0465
20		165.60	-2.29	147.28		0.0155
30		165.60	-0.96	147.28		0.0065
40		165.60	-0.99	147.28		0.0067

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.03	165.60	-11.90	141.08	0.0243	0.0844
20		165.60	-3.44	147.28		0.0234
30	0.11	165.60	-1.51	147.28	0.0006	0.0102
40		165.60	-1.00	147.28		0.0068

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		280.79	-5.53	141.79		0.0390
20	126.59	280.79		147.28	0.4508	
30	52.63	280.79		147.28	0.1874	
40	128.06	280.79		147.28	0.4561	



Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		280.79	-4.72	141.79		0.0333
20	126.43	280.79		147.28	0.4503	
30	52.56	280.79		147.28	0.1872	
40	128.06	280.79		147.28	0.4561	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-4.08	141.08		0.0289
20	128.19	165.60		147.28	0.7741	
30	57.45	165.60		147.28	0.3469	
40	129.81	165.60		147.28	0.7839	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-3.88	141.08		0.0275
20	128.24	165.60		147.28	0.7744	
30	57.46	165.60		147.28	0.3470	
40	129.82	165.60		147.28	0.7839	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-4.12	141.08		0.0292
20	128.18	165.60		147.28	0.7741	
30	57.45	165.60		147.28	0.3469	
40	129.81	165.60		147.28	0.7839	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-4.08	141.08		0.0289
20	128.19	165.60		147.28	0.7741	
30	57.45	165.60		147.28	0.3469	
40	129.81	165.60		147.28	0.7839	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-4.12	141.79		0.0290
20		165.60	-1.81	147.28		0.0123
30		165.60	-0.73	147.28		0.0050
40		165.60	-0.18	147.28		0.0012

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		138.00	-3.94	117.57		0.0335
20	128.26	138.00		122.73	0.9295	
30	57.48	138.00		122.73	0.4166	
40	129.82	138.00		122.73	0.9407	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		138.00	-3.94	117.57		0.0335
20		138.00	-1.73	122.73		0.0141



30	138.00	-0.70	122.73	0.0057
40	138.00	-0.97	122.73	0.0079

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-4.12	141.08		0.0292
20	128.18	165.60		147.28	0.7741	
30	57.45	165.60		147.28	0.3469	
40	129.81	165.60		147.28	0.7839	

Absolute Maximum of the all of the Stress Ratio's 0.9407

Governing Element: Top Dish End
 Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt	...	-3.9	...	...	...	+ Tens
Skirt	...	-3.9	...	...	-3.9	- Comp
Bottom Dish	130.0	-1.7	...	...	128.3	+ Tens
Bottom Dish	130.0	-1.7	...	...	...	- Comp
Main Shell	58.2	-0.7	...	...	57.5	+ Tens
Main Shell	58.2	-0.7	...	...	...	- Comp
Top Dish End	130.0	-0.2	...	...	129.8	+ Tens
Top Dish End	130.0	-0.2	...	...	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt

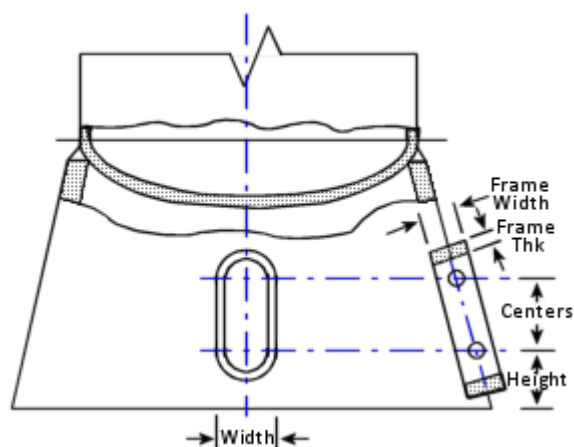
Mean skirt diameter at the critical height is computed later.

This is a Conical type Support Skirt , Wider at the Top

Half Apex Angle of the skirt	a	0.573 deg
Total vessel (empty) weight	Fg	36597.980 Kgf
Head Weight below section 2-2	DeltaFg	5766.032 Kgf
Weight of vessel contents (liquid)	Ff	308.869 Kgf
Vessel Gross Weight	Fg + Ff	36906.852 Kgf
Skirt Mean Diameter Top	DzTop	1336.501 mm.
Skirt Mean Diameter Bottom End	DzBot	1276.501 mm.
Skirt Analysis Thickness (Corroded)	ez	23.500 mm.
Skirt Height	H	3000.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kgf
Moment at section 1-1	M1	11236.164 Kg-m.
Applied Moment - Bottom of Skirt	Mb	23995.541 Kg-m.
Weight at Bottom of Skirt	W	39574.680 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Spherical
Head Mean Diameter at Skirt Top	Db 1349.500 mm.
Head Corroded Thickness	eb 46.500 mm.



Skirt Access Opening Input Data:

Access #	Diameter mm.	Layout Angle deg	Cntr. Dist. From Bottom mm.	Obround Cntr-Cntr mm.	Frame Width mm.	Frame Thickness mm.
1	1200.0000	0.0	1200.00	0.00	0.000	0.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1	Layout Angle 2
-----------	----------------	----------------



	Deg.	Deg.
1	-61.8870	61.8870

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
-298.113	-61.887

Height of Weakest Section	H	1188.858	mm.
Skirt mean diameter at height H	Dz	1360.278	mm.
Rotation of Principle Axes	a	-0.000	deg.
Centroid Position in X direction	g	-291.006	mm.
Centroid Position in Y direction	h	0.000	mm.
Principle Moment of Inertia	Ixx	1831565.500	cm ⁴
Principle Moment of Inertia	Iyy	658867.125	cm ⁴
Distance to Extreme Fiber	ybar	691.889	mm.
Distance to Extreme Fiber	xbar	982.895	mm.
Minimum Section Modulus	W4	6703339.500	mm. ³
Metal Area at the Cutting Plane	A4	659.011	cm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / Dz)$$

$$= \sin^{-1}(1200.0/1360.278)$$

$$= 1.0805 \text{ rad (61.9055 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= Dz / 2$$

$$= 1360.278/2$$

$$= 680.1 \text{ mm.}$$

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / ez)}$$

$$= 1.08 \cdot \sqrt{(680.139/23.5)}$$

$$= 5.813$$

WARNING: ratio Exceeds 2 - See EN 13445 16.12.76(a).
 This equation is a precaution against radial buckling at the opening.
 The frame (if there is one), may prevent this. The code does not say what size this frame should be. Your judgement is required.

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= ex(\text{new}) - ci - cext$$

$$= 25.0 - 1.5 - 0.0$$

$$= 23.5 \text{ mm.}$$

Find the moment acting at the Critical Height [M4]:

$$= (M_{\text{bottom}} - M_{\text{top}}) \cdot (H - \text{CutHeight}) / H + M_{\text{top}}$$

$$= (23995.543 - 11236.165) \cdot (3000.0 - 1188.858) / 3000.0 + 11236.165$$

$$= 18939.178 \text{ Kg-m.}$$

Compute the section modulus [Zxx and Zyy]:

$$Zxx = Ixx / ybar = 1831566 / 691.889 = 26471972 \text{ mm.}^3$$

$$Zyy = Iyy / xbar = 658867 / 982.895 = 6703340 \text{ mm.}^3$$

$$Z = \min(Zxx, Zyy) = \min(26471972, 6703340) = 6703340 \text{ mm.}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(Ixx, Iyy) / \text{Area}(A4))}$$

$$= \sqrt{(\min(1831566, 658867) / 659.011)}$$

$$= 316.193 \text{ mm.}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 316.193/3000.0$$

$$= 0.105$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(e_x^2 + e_y^2)}$$

$$= \sqrt{(-291.006^2 + 0.^2)}$$

$$= 291.006 \text{ mm.}$$

Where $e_x = -291.006$: $e_y = 0$.

Moment Increment at Centroid [ΔM_4]:

$$= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e$$

$$= \text{abs}(0.0 - 36597.98 - 308.87) \cdot 291.006$$

$$= 10740.312 \text{ Kg-m.}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= 0.0 - 36597.98 - 308.869 + 4 \cdot 11236.165/1336.501$$

$$= -3278.9 \text{ Kg}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= 0.0 - 36597.98 - 308.869 - 4 \cdot 11236.165/1336.501$$

$$= -70534.8 \text{ Kg}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-3278.926 + 5766.032 + 308.869) / (\pi \cdot 1349.5 \cdot 46.5) + 191.0 \cdot 1349.5 / (4 \cdot 46.5)$$

$$= 138.72 \text{ N./mm}^2$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-70534.773 + 5766.032 + 308.869) / (\pi \cdot 1349.5 \cdot 46.5) + 191.0 \cdot 1349.5 / (4 \cdot 46.5)$$

$$= 135.38 \text{ N./mm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (308.869 + 5766.032) / (\pi \cdot 1349.5 \cdot 46.5) + 191.0 \cdot 1349.5 / (4 \cdot 46.5)$$

$$= 138.89 \text{ N./mm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= -3278.926 / (\pi \cdot 1336.501 \cdot 23.5)$$

$$= -0.33 \text{ N./mm}^2$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -70534.773 / (\pi \cdot 1336.501 \cdot 23.5)$$

$$= -7.01 \text{ N./mm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$

$$= 1 - 0.5(1349.5 + 46.5 - 1276.5 + 23.5) / (651.5 + 46.5)$$

$$= 0.9405$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.941) = 19.857 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(46.5^2 + 23.5^2 + 2 \cdot 46.5 \cdot 23.5 \cdot 0.941)}$$

$$= 34.533 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 34.53 \cdot -3278.9 = -113.23 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 34.53 \cdot -70534.8 = -2435.81 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057 (eb/ez)^2$$

$$= 0.63 - 0.057 (46.5/23.5)^2$$

$$= 0.407$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.407 \cdot 6 \cdot -113.233 / (\pi \cdot 1349.5 \cdot 46.5^2)$$

$$= -0.30 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.407 \cdot 6 \cdot -2435.813 / (\pi \cdot 1349.5 \cdot 46.5^2)$$

$$= -6.36 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.407 \cdot 6 \cdot -113.233 / (\pi \cdot 1336.501 \cdot 23.5^2)$$

$$= -1.17 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.407 \cdot 6 \cdot -2435.813 / (\pi \cdot 1336.501 \cdot 23.5^2)$$

$$= -25.15 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 1.831/53.13 = 0.0345$$

$$y = 125 \cdot eb/D_b = 125 \cdot 46.5/1349.5 = 4.3072$$

$$\alpha = 0$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= 0.0 \text{ N./mm}^2 \text{ (16.12-31)}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 138.725 - -0.296 + 0.0$$

$$= 139.02 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 138.725 + -0.296 - 0.0$$

$$= 138.43 \text{ N./mm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 135.379 - -6.361 + 0.0$$

$$= 141.74 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 135.379 + -6.361 - 0.0$$

$$= 129.02 \text{ N./mm}^2$$

Stress [σ_{2pitot}]:
 $= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$
 $= 138.888 + -0.296 + 0.0$
 $= 138.59 \text{ N./mm}^2$

Stress [σ_{2potot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 138.888 - -0.296 - 0.0$
 $= 139.18 \text{ N./mm}^2$

Stress [σ_{2qitot}]:
 $= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$
 $= 138.888 + -6.361 + 0.0$
 $= 132.53 \text{ N./mm}^2$

Stress [σ_{2qotot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 138.888 - -6.361 - 0.0$
 $= 145.25 \text{ N./mm}^2$

Stress [σ_{3pitot}]:
 $= \sigma_{3pm} - \sigma_{3pb}$
 $= -0.326 - -1.169$
 $= 0.84 \text{ N./mm}^2$

Stress [σ_{3potot}]:
 $= \sigma_{3pm} + \sigma_{3pb}$
 $= -0.326 + -1.169$
 $= -1.49 \text{ N./mm}^2$

Stress [σ_{3qitot}]:
 $= \sigma_{3qm} - \sigma_{3qb}$
 $= -7.01 - -25.146$
 $= 18.14 \text{ N./mm}^2$

Stress [σ_{3qotot}]:
 $= \sigma_{3qm} + \sigma_{3qb}$
 $= -7.01 + -25.146$
 $= -32.16 \text{ N./mm}^2$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$
 $= 137.997(3 - (1/1.5)) \cdot (138.725/137.997)^2$
 $= 321.02 \text{ N./mm}^2$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$
 $= 137.997(3 - (1/1.5)) \cdot (135.379/137.997)^2$
 $= 325.45 \text{ N./mm}^2$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$
 $= 137.997(3 - (1/1.5)) \cdot (138.888/137.997)^2$
 $= 320.80 \text{ N./mm}^2$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$
 $= 137.997(3 - (1/1.5)) \cdot (138.888/137.997)^2$
 $= 320.80 \text{ N./mm}^2$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2$
 $= 137.997(3 - (1/1.5)) \cdot (-0.326/137.997)^2$
 $= 413.99 \text{ N./mm}^2$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2$
 $= 137.997(3 - (1/1.5)) \cdot (-7.01/137.997)^2$

= 413.76 N./mm²

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:

$$\begin{aligned} &= \text{abs}((lt - ea3)) / 2 \\ &= \text{abs}((0.0 - 23.5)) / 2 \\ &= 11.750 \text{ mm.} \end{aligned}$$

Mean Diameter of Opening, Figure 16.12-5 [d]:

$$\begin{aligned} &= \text{opening width} - et \\ &= 1200.0 - 0.0 \\ &= 1200.000 \text{ mm.} \end{aligned}$$

Angle, 16.12-72 [δ]:

$$\begin{aligned} &= \sin^{-1}(d / D3) \\ &= \sin^{-1}(1200.0 / 1336.501) \\ &= 1.115 \text{ radians} \end{aligned}$$

Area in the Frame [At]:

$$\begin{aligned} &= 2 \cdot lt \cdot et \\ &= 2 \cdot 0.0 \cdot 0.0 \\ &= 0.000 \text{ cm}^2 \end{aligned}$$

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:

$$\begin{aligned} &= (\pi - \delta) \cdot D3 \cdot ea3 \\ &= (\pi - 1.115) \cdot 1336.501 \cdot 23.5 \\ &= 636.539 \text{ cm}^2 \end{aligned}$$

Combined Area at Opening, 16.12-73 [A4]:

$$\begin{aligned} &= AS + AT \\ &= 636.539 + 0.0 \\ &= 636.539 \text{ cm}^2 \end{aligned}$$

Computed Offset, 16.12-74(b) [yt]:

$$\begin{aligned} &= 0.5 \cdot D3 \cdot \cos(\delta) + ht - 0.5 \cdot lt \\ &= 0.5 \cdot 1336.501 \cdot \cos(1.115) + 11.75 - 0.5 \cdot 0.0 \\ &= 305.959 \text{ mm.} \end{aligned}$$

Distance to Neutral Axis, 16.12-74 [yG]:

$$\begin{aligned} &= (0.5 \cdot D3 \cdot ea3 \cdot d - 2 \cdot lt \cdot eat \cdot yt) / A4 \\ &= (0.5 \cdot 1336.501 \cdot 23.5 \cdot 1200.0 - 2 \cdot 0.0 \cdot 0.0 \cdot 305.959) / 636.539 \\ &= 296.049 \text{ mm.} \end{aligned}$$

Computed Offset [ymax]:

$$\begin{aligned} &= \max[\text{abs}(0.5 \cdot d3 \cdot \cos(\delta) + ht + yG) ; \text{abs}(0.5 \cdot D3 - yG)] \\ &= \max[\text{abs}(0.5 \cdot 1336.501 \cdot \cos(1.115) + 11.75 + 296.049) ; \text{abs}(0.5 \cdot 1336.501 - 296.049)] \\ &= 602.008 \text{ mm.} \end{aligned}$$

Inertia Value, 16.12-75(b) [IS]:

$$\begin{aligned} &= (\pi - \delta - \sin(\delta) \cdot \cos(\delta)) \cdot ea3 \cdot (D3/2)^3 \\ &= (\pi - 1.115 - \sin(1.115) \cdot \cos(1.115)) \cdot 23.5 \cdot (1336.501/2)^3 \\ &= 1144046.9 \text{ cm}^4 \end{aligned}$$

Section Modulus [W4]:

$$\begin{aligned} &= (Is + At(yt^2 + lt^2/12)) - A4 \cdot yG^2 / ymax \\ &= (1144047 + 0.0(305.959^2 + 0.0^2/12)) - 636.539 \cdot 296.049^2 / 602.008 \\ &= 9736645.000 \text{ mm}^3 \end{aligned}$$

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned} &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\ &= \min(1; 636.539 / (\pi \cdot 1336.501 \cdot 23.5)) \\ &= 0.645 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned} &= \min(1; 4 \cdot W4 / (\pi \cdot D^2 / 3 / (\pi \cdot D3 \cdot ea3))) \\ &= \min(1; 4.0 \cdot 9736645 / (\pi \cdot 1336.501^2 \cdot 23.5)) \\ &= 0.295 \end{aligned}$$

Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi/4 \cdot d_3^2 \cdot e_{a3} \cdot \sigma_{call} \\ &= \pi/4 \cdot 1336.501^2 \cdot 23.5 \cdot 141.8 \\ &= 476667.9 \text{ Kg-m.} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d_3 \cdot e_{a3} \cdot \sigma_{call} \\ &= \pi \cdot 1336.501 \cdot 23.5 \cdot 141.8 \\ &= 1426585.9 \text{ Kgf} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F_4 / (\psi_{i1} \cdot F_{c,max}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_q)) / (\psi_{i2} \cdot M_{max}) \\ &= 36906.9 / (0.645 \cdot 1426586) + (\text{abs}(18939.2) + 36906.9 \cdot \text{abs}(296.049)) / \\ &\quad (0.295 \cdot 476667.9) \\ &= 0.252 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	139.02	321.02	Pass	46
Stot1po	138.43	321.02	Pass	47
Stot1qi	141.74	325.45	Pass	48
Stot1qo	129.02	325.45	Pass	49
Stot2pi	138.59	320.80	Pass	50
Stot2po	139.18	320.80	Pass	51
Stot2qi	132.53	320.80	Pass	52
Stot2qo	145.25	320.80	Pass	53
Stot3pi	0.84	413.99	Pass	54
Stot3po	-1.49	413.99	Pass	55
Stot3qi	18.14	413.76	Pass	56
Stot3qo	-32.16	413.76	Pass	57

The stresses are satisfactory in the region without openings.
The stresses are satisfactory in the region with the opening(s).

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	5479.993 mm.
Center of Gravity of the Insulation	5479.969 mm.
Center of Gravity of the Nozzles	4007.686 mm.
Center of Gravity of the Added Weights (Operating)	3155.000 mm.
Center of Gravity of the Added Weights (Empty)	3155.000 mm.
Center of Gravity of the Tubesheet(s)	5479.969 mm.
Center of Gravity of the Tubes	5479.969 mm.
Center of Gravity of Bare Shell New and Cold	4955.443 mm.
Center of Gravity of Bare Shell Corroded	4960.274 mm.
Vessel CG in the Operating Condition	4901.233 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	4893.627 mm.
Vessel CG in the Test Condition	4973.087 mm.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	1300.0001	mm.
Skirt Thickness	STHK	25.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.5000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material		SA-516 70	[Impact Tested]

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	32.0000	mm.
Design Temperature of the Basing		85.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	N./mm ²
Basing Yield Stress		242.00	N./mm ²
Inside Diameter of Basing	DI, D4	1246.0000	mm.
Outside Diameter of Basing	DOU, D5	1552.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	27.0000	mm.
Bolt Corrosion Allowance	BCA	0.0000	mm.
Bolt Material		SA-193 B8	
Bolt Material Class/Thickness Range		1::	
Bolt Operating Allowable Stress	SA, fB	117.43	N./mm ²
Number of Bolts	RN, nb	12	
Diameter of Bolt Circle	DC, DBC	1462.0000	mm.
Thickness of Gusset Plates	TGA, e7	10.0000	mm.
Width of Gussets at Top Plate	TWDT	96.0000	mm.
Width of Gussets at Base Plate	BWDT	96.0000	mm.
Gusset Plate Elastic Modulus	E	199042000.0	KPa.
Gusset Plate Yield Stress	SY	242.0	N./mm ²
Height of Gussets	HG, h1	117.0000	mm.
Distance between Gussets	RG, b6	78.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	39.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	24.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	117.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	67.0000	mm.
External Corrosion Allowance	CA	1.5000	mm.
Dead Weight of Vessel	DW	39206.1	Kgf
Operating Weight of Vessel	ROW	39574.7	Kgf
Earthquake Moment on Basing	EQMOM	23995.5	Kg-m.
Wind Moment on Basing	WIMOM	7368.9	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Basing Analysis : Analyze Option

Basing Thickness Calculation method used : Simplified (Steel on Steel)

Calculation of Bolt Loads for Multiple Loading Conditions:

Bolt Load, Earthquake Operating Case [FB,e]:

$$= ((4 \cdot M/DC) - \text{abs}(W - \text{vertical force}))/nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 23996/1462.0) - 37855)/12$$

$$= 2316.3 \text{ Kgf}$$

Bolt Load, Windload Empty Case [FB,w]:

$$= ((4 \cdot M/DC) - W)/nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 7369/1462.0) - 39206)/12$$

$$= -1587.1 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case [FB,v]:

$$= ((4 \cdot M/DC) - W)/nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/1462.0) - 39575)/12$$

$$= -3297.9 \text{ Kgf}$$

Calculation of Maximum Load per Bolt [W/Bolt], Earthquake + Operating Condition:
 where: W = ROW - Yforce (from gy Accelration or user force), M = EQMOM

$$= ((4 \cdot M/DC) - W)/nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 23996/1462.0) - 37855)/12$$

$$= 2316.3 \text{ Kgf}$$

Required Area for Each Bolt, Based on Max Load	1.9344	cm ²
Area Available in a Single Bolt (Corr)	4.1385	cm ²
Area Available in all the Bolts (Corr)	49.6622	cm ²
Bolt Stress Based on Simplified Analysis	54.9	N./mm ²
Allowable Bolt Stress 117.4 [Fact]	117.43	N./mm ²

Concrete Contact Area of Base Ring	CCA	6724.49	cm ²
Concrete Contact Section Modulus of Base Ring		0.2145E+09	mm. ³

Concrete Load (Simplified method), Earthquake in Operating Condition [Sc]:

$$= ((ppl/100 \cdot (Abt \cdot Sa) + W)/Cca) + M/CZ \text{ per Jawad \& Farr Eq. 12.1}$$

$$= (1.0(49.6622 \cdot 117 + 41295)/6724.49) + 23996/21454E+09$$

$$= 2.57 \text{ N./mm}^2$$

Concrete Allowable Stress	10.00	N./mm ²
---------------------------	-------	--------------------

Determine Maximum Bending Width of Basing Section [Rw1]:

$$= (Dou - SkirtOD)/2$$

$$= (1552.0 - 1300.0)/2$$

$$= 126.0 \text{ mm.}$$

Determine Maximum Bending Width of Basing Section [Rw2]:

$$= (SkirtID - Di + 2 \cdot Sca)/2$$

$$= (1250.0 - 1246.0 + 2 \cdot 1.5)/2$$

$$= 3.5 \text{ mm.}$$

Calculation of required Basing Thickness, (Simplified) [Tb]:

$$= \max(Rw1, Rw2) \cdot \sqrt{(3 \cdot Sc / S)} + CA \text{ per Jawad \& Farr Eq. 12.12}$$

$$= \max(126.0, 3.5) \cdot \sqrt{(3 \cdot 2.566/161.331)} + 1.5$$

$$= 29.0250 \text{ mm.}$$

Basing Stress at given Thickness [Sb]:

$$= 3 \cdot Sc(\max[Rw1, Rw2]/(Tb - Ca))^2$$

$$= 3 \cdot 2.566(\max[126.0, 3.5]/(32.0 - 1.5))^2$$

$$= 131.393, \text{ must be less than } 161.331 \text{ N./mm}^2$$

Required Thickness of Top Plate in Tension:
 (Calculated as a fixed beam per Megyesy)

$$Ft = (Sa \cdot Abss), \text{ Bolt Allowable Stress} \cdot \text{Area}$$

$$Rm = (Ft \cdot 2 \cdot Cg)/8, \text{ Bending Moment}$$

$$Sb = \text{Allowable Bending Stress}$$

$$Wt = (Topwth - Bnd), \text{ Width of Section}$$

Required Thickness of Top Plate [T]:

$$= \sqrt{(6 \cdot Rm / (Sb \cdot Wt))} + CA$$

$$= \sqrt{(6 \cdot 48 / (161 \cdot 90.0))} + 1.5$$

$$= 15.493 \text{ mm.}$$

Required Thickness of Continuous Top Ring per Moss:

$$a = (Dc - SkirtOD)/2 \text{ Skirt Distance to Bolt Circle}$$

$$P = Sa \cdot Abss \text{ Bolt Allowable Stress} \cdot \text{Area}$$

$$l = Avgwdt \text{ Average Gusset Width}$$

$$g1 = \text{Gamma } 1 \text{ Constant Term } f(b/l)$$

$$g2 = \text{Gamma } 2 \text{ Constant Term } f(b/l)$$

$$g = \text{Flat distance} / 2 \text{ Nut } 1/2 \text{ Dimension (from Tema)}$$

$$Fb = \text{Allowable Bending Stress}$$

$$Mo = P/(4\pi)[1.3(\ln((2l\sin(\pi a/l)/(\pi g)))+1)] - [(0.7-g2)P/(4\pi)] \text{ Moment Term}$$

Required Thickness of Top Ring [T]:
 $= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb})} + \text{CA}$ Required Thickness
 $= \sqrt{(6 \cdot 7/161)} + 1.5$
 $= 11.513 \text{ mm.}$

Required Thickness of Gusset in Compression, per AISC E2-1:

1. Allowed Compression at Given Thickness:

Factor K1/r Per E2-1	40.5293	
Factor Cc Per E2-1	127.4221	
Allowable Buckling Str. per E2-1	128.94	N./mm ²
Actual Buckling Str. at Given Thickness	25.31	N./mm ²

Required Gusset/Support Plate thickness, + CA 4.9701 mm.

2. Allowed Compression at Calculated Thickness:

Factor K1/r Per E2-1	116.7944	
Factor Cc Per E2-1	127.4221	
Allowable Buckling Str. per E2-1	73.32	N./mm ²
Act. Buckling Str. at Calculated Thickness	72.94	N./mm ²

Summary of Basing Thickness Calculations:

Required Basing Thickness (simplified)	29.0250	mm.
Actual Basing Thickness	32.0000	mm.

Required Top Ring/Plate Thickness as a Fixed Beam	15.4928	mm.
Required Thickness of Continuous Top Ring (Moss)	11.5134	mm.
Actual Top Plate/Ring Thickness	24.0000	mm.

Required Gusset/Support Plate thickness, + CA	4.9701	mm.
Actual Gusset/Support Plate Thickness	10.0000	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:
 $= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$
 $= 11.905 \cdot (25.0 - 1.5)$
 $= 28.53 \text{ Kg/mm.}$

Results for Computed Minimum Basing Weld Size [BWeld]:
 $= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 29 / [(0.4 \cdot 242) \cdot 2 \cdot 0.707]$
 $= 2.044 \text{ mm.}$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:
 $= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$
 $= 4955.5 / (98.0 + 2 \cdot (117.0 + 24.0))$
 $= 13.041 \text{ Kg/mm.}$

Horizontal Plate Load [Wh]:
 $= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$
 $= 4955.5 \cdot 81.0 / (98.0 \cdot (141.0) + 0.6667 \cdot (141.0)^2)$
 $= 14.827 \text{ Kg/mm.}$

Resultant Weld Load [Wr]:
 $= \sqrt{Wv^2 + Wh^2}$
 $= \sqrt{(13.04^2 + 14.83^2)}$
 $= 19.746 \text{ Kg/mm.}$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:
 $= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 19.75 / [(0.4 \cdot 242) \cdot 2 \cdot 0.707]$
 $= 1.415 \text{ mm.}$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
 $= \text{Bolt Load} / (2 \cdot \text{TopWth})$

$$= 4955.5 / (2 \cdot 96.0)$$
$$= 25.81 \text{ Kg/mm.}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$
$$= 4955.5 \cdot 81.0 / (2 \cdot 141.0 \cdot 96.0)$$
$$= 14.827 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$
$$= \sqrt{25.81^2 + 14.83^2}$$
$$= 29.766 \text{ Kg/mm.}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 29.77 / [(0.4 \cdot 242) \cdot 2 \cdot 0.707]$$
$$= 2.133 \text{ mm.}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm.
Required Gusset to Skirt Double Fillet Weld Size	4.7625	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	2.1327	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: B1-Tube Outlet

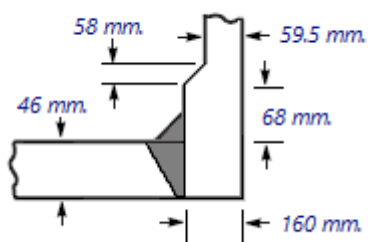
From: 20

Pressure for Reinforcement Calculations	P	191.037	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Hemispherical Head	D	1300.00	mm.
Head Finished (Minimum) Thickness	t	48.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-20.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	67.9834	mm.
Height of Beveled Transition	L'	58.0166	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	46.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1200.0	mm.
Centerline Distance to Tangent		2000.0	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: B1-Tube Outlet



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 490.576 mm.
Actual Thickness Used in Calculation 59.512 mm.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $191.04 \cdot 651.5 / (2 \cdot 138 \cdot 1.0 - 0.2 \cdot 191.04)$
= 45.7309 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.04 \cdot 246.79 / (138 \cdot 1.0 - 0.6 \cdot 191.04)$
= 37.2611 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $100.488 / 0.5773$
= 174.0500 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

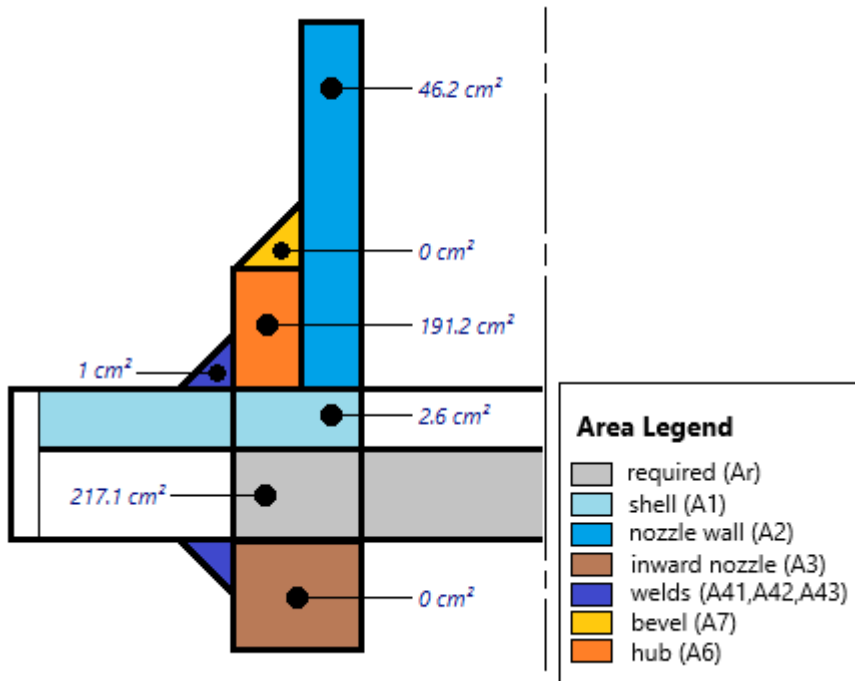
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	987.1512	mm.
Parallel to Vessel Wall, opening length	d	493.5756	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	116.2500	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	225.716	NA	NA
Area in Shell	A1	3.796	NA	NA
Area in Nozzle Wall	A2	48.246	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	193.283	NA	NA
TOTAL AREA AVAILABLE	Atot	246.326	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (493.5756 \cdot 45.7309 \cdot 1.0 + 2 \cdot 158.5 \cdot 45.7309 \cdot 1.0 \cdot (1 - 1.0))$
 $= 225.716 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 493.576 (1.0 \cdot 46.5 - 1.0 \cdot 45.731) - 2 \cdot 158.5$
 $(1.0 \cdot 46.5 - 1.0 \cdot 45.7309) \cdot (1 - 1.0)$
 $= 3.796 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 116.25) (58.01 - 37.26) 1.0$
 $= 48.246 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(116.2, 200.0, 68.0)) \cdot (160.0 - 59.5) \cdot 1.0$$

$$= 193.283 \text{ cm}^2 \text{ Includes Bevel Area of } 56.653 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 38.7611 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 47.2309 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 47.2309 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 9.8312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.831, \max(47.2309, 3.0)]$$

$$= 9.8312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(38.7611, 9.8312)$$

$$= 38.7611 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073 \text{ mm.} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 48.0, tr = 45.731, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.983$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	15 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 15 °C

Weld Size Calculations, Description: B1-Tube Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (225.7164 - 3.7963 + 2 \cdot 158.5 \cdot 1.0 \cdot (1.0 \cdot 46.5 - 45.7309)) \cdot 138)$$

$$= 315707.25 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an} / 707)^2 \cdot fr_2) \cdot S_v$$

$$= (241.5294 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 341276.91 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (241.5294 + 0.0 + 1. + (147.405)) \cdot 138$$

$$= 548698.88 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (241.5294 + 0.0 + 1. + 0.0 + (147.405)) \cdot 138$$



= 548698.88 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 810.5756 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 87791. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 326.0378) \cdot (160.0 - 1.5) \cdot 0.7 \cdot 138 \\ &= 1599144. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 810.5756 \cdot (46.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 609882. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (87791 + 1599144) = 1686936 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (87791 + 0 + 609882 + 0) = 697673 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (87791 + 609882 + 0) = 697673 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1686935 Kgf, must exceed W = 315707 Kgf or W1 = 341276 Kgf
Path 2-2 = 697672 Kgf, must exceed W = 315707 Kgf or W2 = 548698 Kgf
Path 3-3 = 697672 Kgf, must exceed W = 315707 Kgf or W3 = 548698 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 194 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 141.8250 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 389.8250 mm.

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Input, Nozzle Desc: B2-Shell outlet

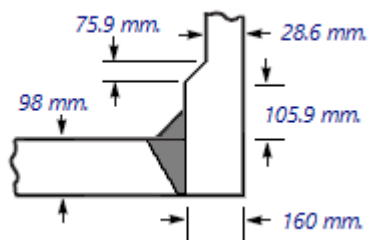
From: 30

Pressure for Reinforcement Calculations	P	191.004	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1250.00	mm.
Shell Finished (Minimum) Thickness	t	98.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4055.00	mm.
User Entered Minimum Design Metal Temperature		-20.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		250.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	105.8780	mm.
Height of Beveled Transition	L'	75.8782	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	98.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: B2-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	215.900	mm.
Actual Thickness Used in Calculation	28.575	mm.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.0 \cdot 626.5 / (138 \cdot 1.0 - 0.6 \cdot 191.0)$
= 94.5744 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.0 \cdot 109.45 / (138 \cdot 1.0 - 0.6 \cdot 191.0)$
= 16.5222 mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $96.5 / 0.5773$
= 167.1430 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

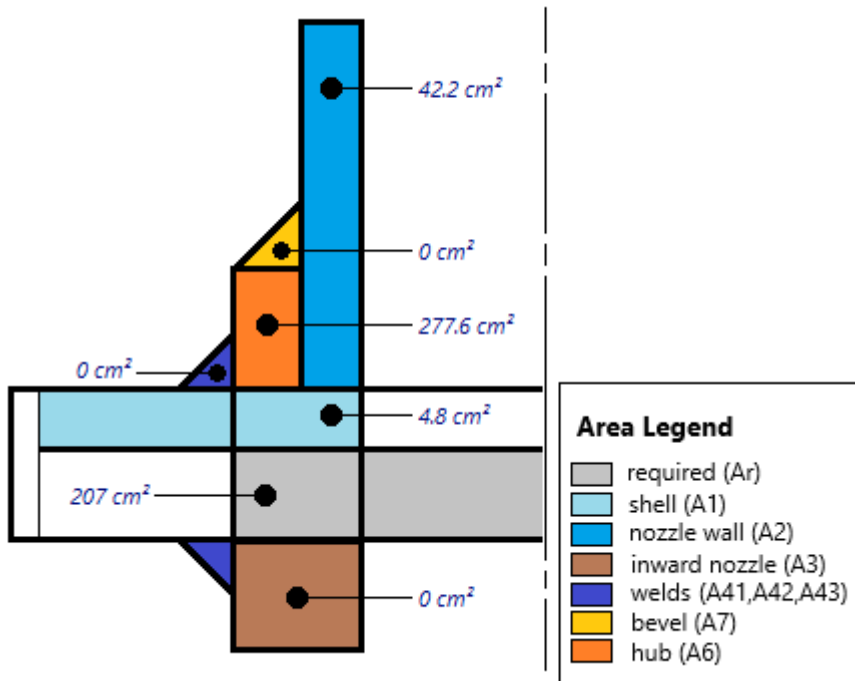
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	466.0500	mm.
Parallel to Vessel Wall	Rn+tn+t	233.0250	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	234.8305	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	207.023	NA	NA
Area in Shell	A1	4.759	NA	NA
Area in Nozzle Wall	A2	42.211	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	277.567	NA	NA
TOTAL AREA AVAILABLE	Atot	324.537	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (218.9 \cdot 94.5744 \cdot 1.0 + 2 \cdot 123.575 \cdot 94.5744 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 207.023 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 247.15 (1.0 \cdot 96.5 - 1.0 \cdot 94.574) - 2 \cdot 123.575$$

$$(1.0 \cdot 96.5 - 1.0 \cdot 94.5744) \cdot (1 - 1.0)$$

$$= 4.759 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot ho) (tn - trn) fr2$$

$$= (2 \cdot 200.0) (27.08 - 16.52) 1.0$$

$$= 42.211 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can / 0.707)^2 - \text{Area Lost} \cdot fr2$$

$$= (10.0^2 - 1.) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= 0.000 \text{ cm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(200.0, 200.0, 105.9)) \cdot (125.1 - 28.6) \cdot 1.0$$

$$= 277.567 \text{ cm}^2 \text{ Includes Bevel Area of } 73.222 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 18.0222 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 96.0744 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 96.0744 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 3.0000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 9.6026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[9.603, \max(96.0744, 3.0)]$$

$$= 9.6026 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(18.0222, 9.6026)$$

$$= 18.0222 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 28.575 = 25.003 \text{ mm.} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 98.0$, $tr = 94.574$, $c = 1.5 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = 1°C

Min Metal Temp. w/o impact per UCS-66, Curve B	31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	30 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 30 °C

Weld Size Calculations, Description: B2-Shell outlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (207.0234 - 4.7591 + 2 \cdot 123.575 \cdot 1.0 \cdot (1.0 \cdot 96.5 - 94.5744)) \cdot 138)$$

$$= 291314.41 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an} / 707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (319.7781 + 0.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 449978.03 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (319.7781 + 0.0 + 0.0 + (238.4997)) \cdot 138$$

$$= 785584.62 \text{ Kg}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (319.7781 + 0.0 + 0.0 + 0.0 + (238.4997)) \cdot 138$$

= 785584.62 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 466.05 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 50477. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 171.2375) \cdot (125.075 - 1.5) \cdot 0.7 \cdot 138 \\ &= 654817. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 466.05 \cdot (98.0 - 1.5) \cdot 0.74 \cdot 138 \\ &= 735621. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (50477 + 654817) = 705294 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (50477 + 0 + 735621 + 0) = 786098 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (50477 + 735621 + 0) = 786098 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 705293 Kgf, must exceed W = 291314 Kgf or W1 = 449978 Kgf
Path 2-2 = 786097 Kgf, must exceed W = 291314 Kgf or W2 = 785584 Kgf
Path 3-3 = 786097 Kgf, must exceed W = 291314 Kgf or W3 = 785584 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 195 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 45.0652 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 343.0652 mm.

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Input, Nozzle Desc: B3-Shell outlet

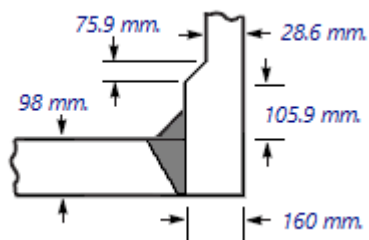
From: 30

Pressure for Reinforcement Calculations	P	191.004	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1250.00	mm.
Shell Finished (Minimum) Thickness	t	98.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4055.00	mm.
User Entered Minimum Design Metal Temperature		-20.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		360.00	deg
Diameter		250.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	105.8780	mm.
Height of Beveled Transition	L'	75.8782	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	98.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: B3-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	215.900	mm.
Actual Thickness Used in Calculation	28.575	mm.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.0 \cdot 626.5 / (138 \cdot 1.0 - 0.6 \cdot 191.0)$
= 94.5744 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.0 \cdot 109.45 / (138 \cdot 1.0 - 0.6 \cdot 191.0)$
= 16.5222 mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 96.5/0.5773
= 167.1430 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

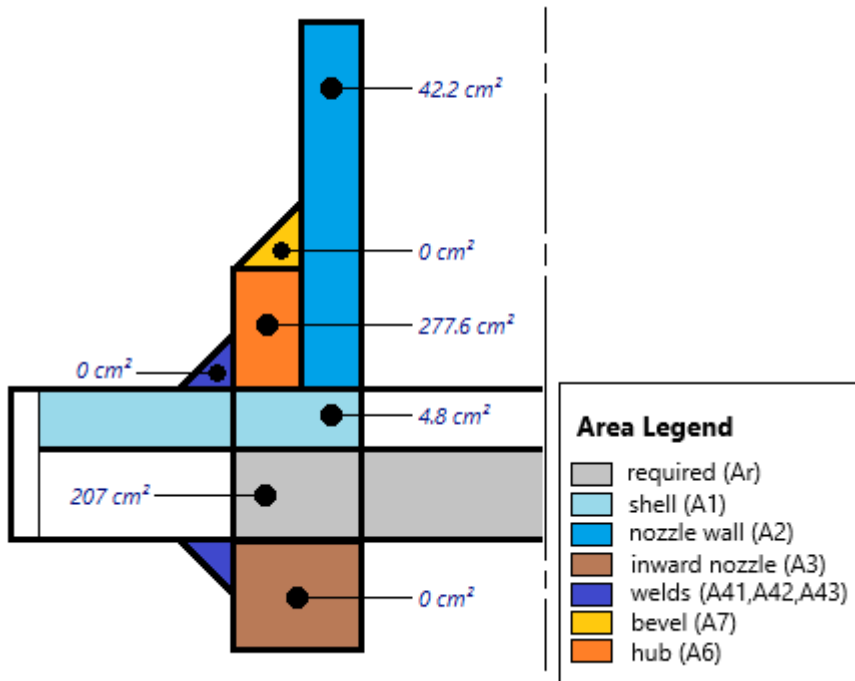
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	466.0500	mm.
Parallel to Vessel Wall	Rn+tn+t	233.0250	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	234.8305	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	207.023	NA	NA
Area in Shell	A1	4.759	NA	NA
Area in Nozzle Wall	A2	42.211	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	277.567	NA	NA
TOTAL AREA AVAILABLE	Atot	324.537	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (218.9 \cdot 94.5744 \cdot 1.0 + 2 \cdot 123.575 \cdot 94.5744 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 207.023 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 247.15 (1.0 \cdot 96.5 - 1.0 \cdot 94.574) - 2 \cdot 123.575$$

$$(1.0 \cdot 96.5 - 1.0 \cdot 94.5744) \cdot (1 - 1.0)$$

$$= 4.759 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot ho) (tn - trn) \cdot fr2$$

$$= (2 \cdot 200.0) (27.08 - 16.52) \cdot 1.0$$

$$= 42.211 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can / 0.707)^2 - \text{Area Lost} \cdot fr2$$

$$= (10.0^2 - 1.) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= 0.000 \text{ cm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(200.0, 200.0, 105.9)) \cdot (125.1 - 28.6) \cdot 1.0$$

$$= 277.567 \text{ cm}^2 \text{ Includes Bevel Area of } 73.222 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$ta = 18.0222 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 96.0744 \text{ mm.}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 96.0744 \text{ mm.}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 9.6026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[9.603, \max(96.0744, 3.0)]$$

$$= 9.6026 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(18.0222, 9.6026)$$

$$= 18.0222 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 28.575 = 25.003 \text{ mm.} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 98.0$, $tr = 94.574$, $c = 1.5 \text{ mm.}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.98$, Temp. Reduction = 1°C

Min Metal Temp. w/o impact per UCS-66, Curve B	31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	30 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 30 °C

Weld Size Calculations, Description: B3-Shell outlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (207.0234 - 4.7591 + 2 \cdot 123.575 \cdot 1.0 \cdot (1.0 \cdot 96.5 - 94.5744)) \cdot 138)$$

$$= 291314.41 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an} / 707)^2 \cdot fr_2) \cdot S_v$$

$$= (319.7781 + 0.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 449978.03 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (319.7781 + 0.0 + 0.0 + (238.4997)) \cdot 138$$

$$= 785584.62 \text{ Kg}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (319.7781 + 0.0 + 0.0 + 0.0 + (238.4997)) \cdot 138$$

= 785584.62 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 466.05 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 50477. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 171.2375) \cdot (125.075 - 1.5) \cdot 0.7 \cdot 138 \\ &= 654817. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 466.05 \cdot (98.0 - 1.5) \cdot 0.74 \cdot 138 \\ &= 735621. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (50477 + 654817) = 705294 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (50477 + 0 + 735621 + 0) = 786098 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (50477 + 735621 + 0) = 786098 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 705293 Kgf, must exceed W = 291314 Kgf	or W1 = 449978 Kgf
Path 2-2 = 786097 Kgf, must exceed W = 291314 Kgf	or W2 = 785584 Kgf
Path 3-3 = 786097 Kgf, must exceed W = 291314 Kgf	or W3 = 785584 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 195 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 45.0652 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 343.0652 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 233.02 mm.

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Input, Nozzle Desc: A2-Shell inlet

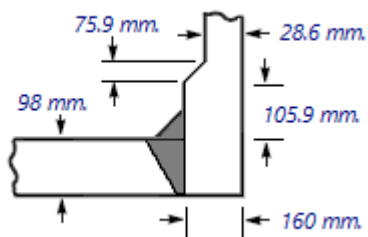
From: 30

Pressure for Reinforcement Calculations	P	191.023	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1250.00	mm.
Shell Finished (Minimum) Thickness	t	98.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		855.00	mm.
User Entered Minimum Design Metal Temperature		-20.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		360.00	deg
Diameter		250.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	105.8780	mm.
Height of Beveled Transition	L'	75.8782	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	98.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: A2-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	215.900	mm.
Actual Thickness Used in Calculation	28.575	mm.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.02 \cdot 626.5 / (138 \cdot 1.0 - 0.6 \cdot 191.02)$
= 94.5846 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.02 \cdot 109.45 / (138 \cdot 1.0 - 0.6 \cdot 191.02)$
= 16.5240 mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $96.5 / 0.5773$
= 167.1430 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

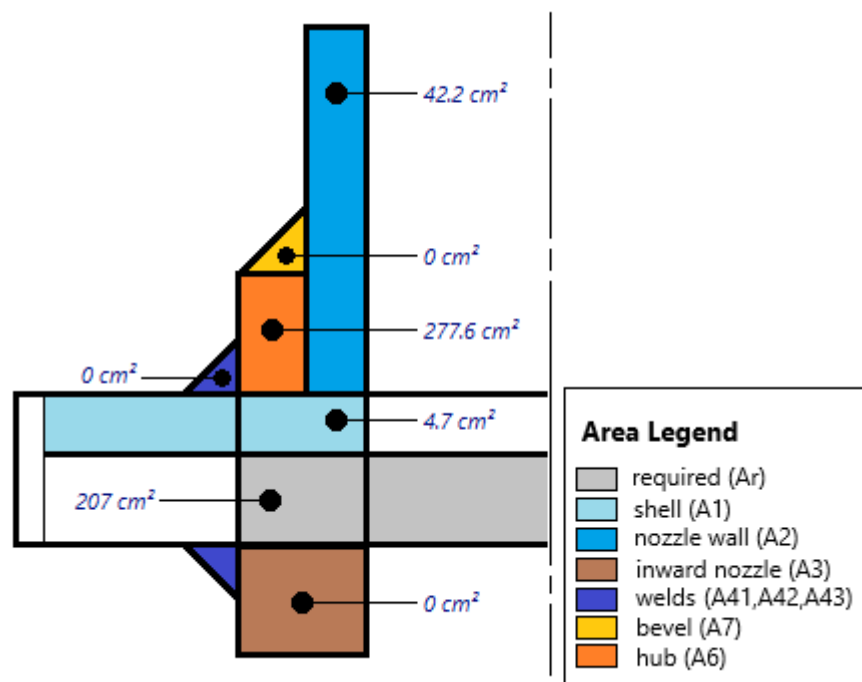
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	466.0500	mm.
Parallel to Vessel Wall	Rn+tn+t	233.0250	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	234.8305	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	207.046	NA	NA
Area in Shell	A1	4.734	NA	NA
Area in Nozzle Wall	A2	42.204	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	277.567	NA	NA
TOTAL AREA AVAILABLE	Atot	324.505	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (218.9 \cdot 94.5846 \cdot 1.0 + 2 \cdot 123.575 \cdot 94.5846 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 207.046 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 247.15 (1.0 \cdot 96.5 - 1.0 \cdot 94.585) - 2 \cdot 123.575$$

$$(1.0 \cdot 96.5 - 1.0 \cdot 94.5846) \cdot (1 - 1.0)$$

$$= 4.734 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot ho) (tn - trn) fr2$$

$$= (2 \cdot 200.0) (27.08 - 16.52) 1.0$$

$$= 42.204 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can / 0.707)^2 - \text{Area Lost} \cdot fr2$$

$$= (10.0^2 - 1.) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= 0.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(200.0, 200.0, 105.9)) \cdot (125.1 - 28.6) \cdot 1.0 \\ &= 277.567 \text{ cm}^2 \text{ Includes Bevel Area of } 73.222 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad t_a = 18.0240 \text{ mm.} \\ \text{Wall Thickness per UG16(b),} & \quad t_{r16b} = 3.0000 \text{ mm.} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad t_{rb1} = 96.0846 \text{ mm.} \\ \text{Wall Thickness} & \quad t_{b1} = \max(t_{rb1}, t_{r16b}) = 96.0846 \text{ mm.} \\ \text{Wall Thickness} & \quad t_{b2} = \max(t_{rb2}, t_{r16b}) = 3.0000 \text{ mm.} \\ \text{Wall Thickness per table UG-45} & \quad t_{b3} = 9.6026 \text{ mm.} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[t_{b3}, \max(t_{b1}, t_{b2})] \\ &= \min[9.603, \max(96.0846, 3.0)] \\ &= 9.6026 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(18.024, 9.6026) \\ &= 18.0240 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 28.575 = 25.003 \text{ mm.} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 98.0$, $t_r = 94.585$, $c = 1.5 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = 1°C

Min Metal Temp. w/o impact per UCS-66, Curve B	31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	30 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 30 °C

Weld Size Calculations, Description: A2-Shell inlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (207.0457 - 4.734 + 2 \cdot 123.575 \cdot 1.0 \cdot (1.0 \cdot 96.5 - 94.5846)) \cdot 138) \\ &= 291345.75 \text{ Kg} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A4 - (W_i - C_{an} / 707)^2 \cdot f_{r2}) \cdot S_v \\ &= (319.771 + 0.0 - 0.0 \cdot 1.0) \cdot 138 \\ &= 449968.00 \text{ Kg} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (319.771 + 0.0 + 0.0 + (238.4997)) \cdot 138 \\ &= 785574.56 \text{ Kg} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (319.771 + 0.0 + 0.0 + 0.0 + (238.4997)) \cdot 138 \end{aligned}$$



= 785574.56 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 466.05 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 50477. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 171.2375) \cdot (125.075 - 1.5) \cdot 0.7 \cdot 138 \\ &= 654817. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 466.05 \cdot (98.0 - 1.5) \cdot 0.74 \cdot 138 \\ &= 735621. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (50477 + 654817) = 705294 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (50477 + 0 + 735621 + 0) = 786098 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (50477 + 735621 + 0) = 786098 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 705293 Kgf, must exceed W = 291345 Kgf or W1 = 449968 Kgf
Path 2-2 = 786097 Kgf, must exceed W = 291345 Kgf or W2 = 785574 Kgf
Path 3-3 = 786097 Kgf, must exceed W = 291345 Kgf or W3 = 785574 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 195 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 45.0652 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 343.0652 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 233.02 mm.

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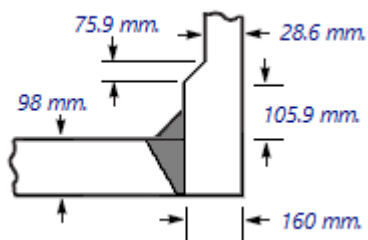
Input, Nozzle Desc: A3-Shell Inlet From: 30

Pressure for Reinforcement Calculations	P	191.023	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1250.00	mm.
Shell Finished (Minimum) Thickness	t	98.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		855.00	mm.
User Entered Minimum Design Metal Temperature		-20.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		250.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	105.8780	mm.
Height of Beveled Transition	L'	75.8782	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	98.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: A3-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	215.900	mm.
Actual Thickness Used in Calculation	28.575	mm.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.02 \cdot 626.5 / (138 \cdot 1.0 - 0.6 \cdot 191.02)$
= 94.5846 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.02 \cdot 109.45 / (138 \cdot 1.0 - 0.6 \cdot 191.02)$
= 16.5240 mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $96.5 / 0.5773$
= 167.1430 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

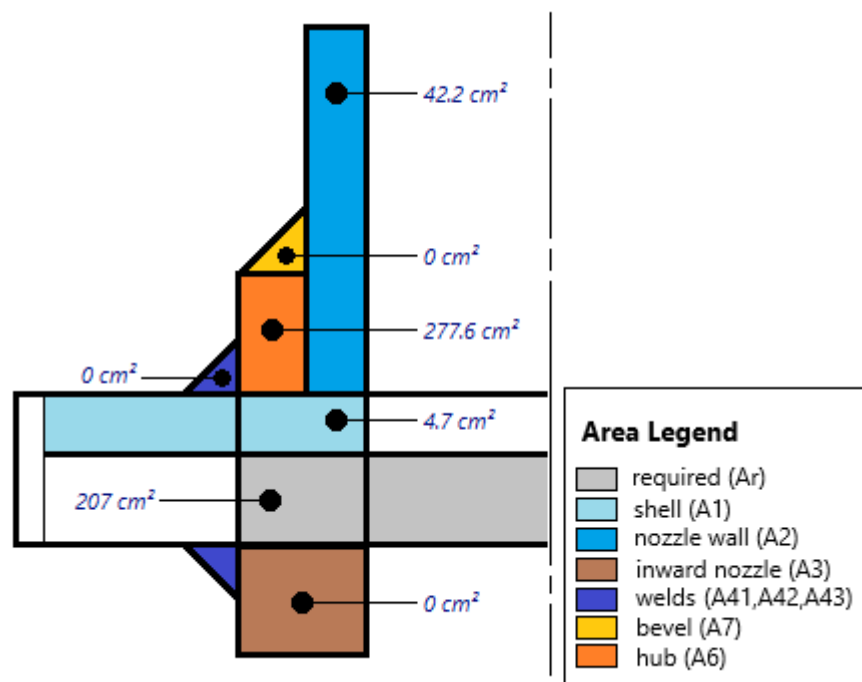
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	466.0500	mm.
Parallel to Vessel Wall	Rn+tn+t	233.0250	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	234.8305	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	207.046	NA	NA
Area in Shell	A1	4.734	NA	NA
Area in Nozzle Wall	A2	42.204	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	277.567	NA	NA
TOTAL AREA AVAILABLE	Atot	324.505	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (218.9 \cdot 94.5846 \cdot 1.0 + 2 \cdot 123.575 \cdot 94.5846 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 207.046 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 247.15 (1.0 \cdot 96.5 - 1.0 \cdot 94.585) - 2 \cdot 123.575$$

$$(1.0 \cdot 96.5 - 1.0 \cdot 94.5846) \cdot (1 - 1.0)$$

$$= 4.734 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot ho) (tn - trn) \cdot fr2$$

$$= (2 \cdot 200.0) (27.08 - 16.52) \cdot 1.0$$

$$= 42.204 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can / 0.707)^2 - \text{Area Lost} \cdot fr2$$

$$= (10.0^2 - 1.) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= 0.000 \text{ cm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(200.0, 200.0, 105.9)) \cdot (125.1 - 28.6) \cdot 1.0$$

$$= 277.567 \text{ cm}^2 \text{ Includes Bevel Area of } 73.222 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 18.0240 \text{ mm.}$
 Wall Thickness per UG16(b), $t_{r16b} = 3.0000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $t_{rb1} = 96.0846 \text{ mm.}$
 Wall Thickness $t_{b1} = \max(t_{rb1}, t_{r16b}) = 96.0846 \text{ mm.}$
 Wall Thickness $t_{b2} = \max(t_{rb2}, t_{r16b}) = 3.0000 \text{ mm.}$
 Wall Thickness per table UG-45 $t_{b3} = 9.6026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[9.603, \max(96.0846, 3.0)]$$

$$= 9.6026 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(18.024, 9.6026)$$

$$= 18.0240 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 28.575 = 25.003 \text{ mm.} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govern. thk, $t_g = 98.0$, $t_r = 94.585$, $c = 1.5 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = 1°C

Min Metal Temp. w/o impact per UCS-66, Curve B 31°C
 Min Metal Temp. at Required thickness (UCS 66.1) 30°C

Gov. MDMT of the nozzle to shell joint welded assembly : 30°C

Weld Size Calculations, Description: A3-Shell Inlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (207.0457 - 4.734 + 2 \cdot 123.575 \cdot 1.0 \cdot (1.0 \cdot 96.5 - 94.5846)) \cdot 138)$$

$$= 291345.75 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an} / 707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (319.771 + 0.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 449968.00 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (319.771 + 0.0 + 0.0 + (238.4997)) \cdot 138$$

$$= 785574.56 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (319.771 + 0.0 + 0.0 + 0.0 + (238.4997)) \cdot 138$$



= 785574.56 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 466.05 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 50477. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 171.2375) \cdot (125.075 - 1.5) \cdot 0.7 \cdot 138 \\ &= 654817. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 466.05 \cdot (98.0 - 1.5) \cdot 0.74 \cdot 138 \\ &= 735621. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (50477 + 654817) = 705294 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (50477 + 0 + 735621 + 0) = 786098 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (50477 + 735621 + 0) = 786098 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 705293 Kgf, must exceed W = 291345 Kgf or W1 = 449968 Kgf
Path 2-2 = 786097 Kgf, must exceed W = 291345 Kgf or W2 = 785574 Kgf
Path 3-3 = 786097 Kgf, must exceed W = 291345 Kgf or W3 = 785574 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 195 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 45.0652 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 343.0652 mm.

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Input, Nozzle Desc: A1-Tube inlet

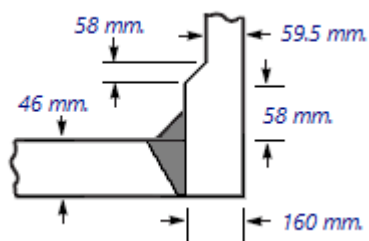
From: 40

Pressure for Reinforcement Calculations	P	191.000	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Hemispherical Head	D	1300.00	mm.
Head Finished (Minimum) Thickness	t	48.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-20.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	58.0166	mm.
Height of Beveled Transition	L'	58.0166	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	46.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: A1-Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	490.576	mm.
--	---------	-----

Actual Thickness Used in Calculation 59.512 mm.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $191.0 \cdot 651.5 / (2 \cdot 138 \cdot 1.0 - 0.2 \cdot 191.0)$
= 45.7220 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191.0 \cdot 246.79 / (138 \cdot 1.0 - 0.6 \cdot 191.0)$
= 37.2533 mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.2308 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $100.488 / 0.5773$
= 174.0500 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

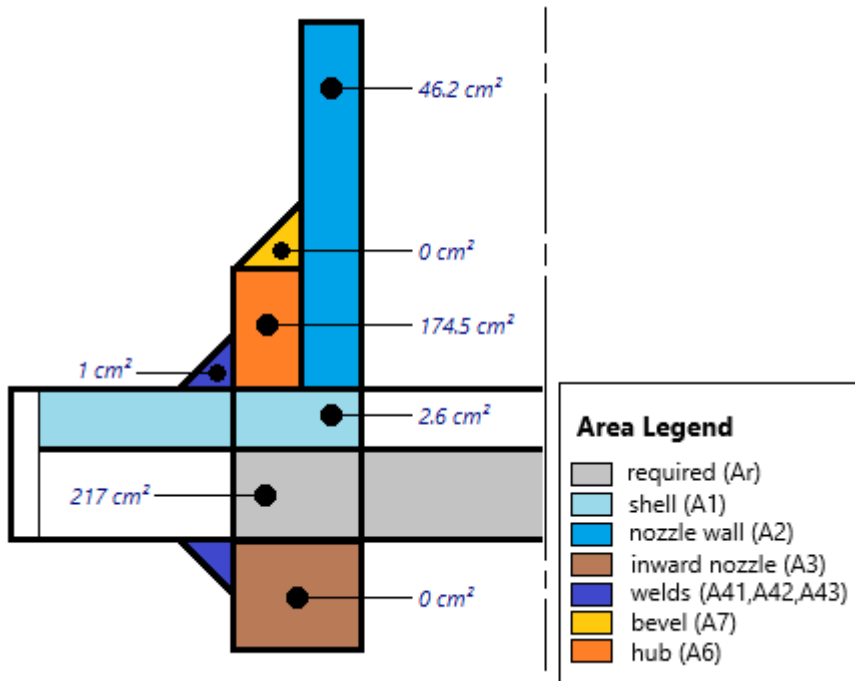
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	987.1512 mm.
Parallel to Vessel Wall, opening length	d	493.5756 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	116.2500 mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	225.673	4.956	NA
Area in Shell	A1	3.840	219.601	NA
Area in Nozzle Wall	A2	48.264	132.017	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	174.899	174.899	NA
TOTAL AREA AVAILABLE	Atot	228.003	527.517	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (493.5756 \cdot 45.722 \cdot 1.0 + 2 \cdot 158.5 \cdot 45.722 \cdot 1.0 \cdot (1 - 1.0))$
 $= 225.673 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 493.576 (1.0 \cdot 46.5 - 1.0 \cdot 45.722) - 2 \cdot 158.5$
 $(1.0 \cdot 46.5 - 1.0 \cdot 45.722) \cdot (1 - 1.0)$
 $= 3.840 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 116.25) (58.01 - 37.25) 1.0$
 $= 48.264 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(116.2, 200.0, 58.0)) \cdot (160.0 - 59.5) \cdot 1.0 \\ &= 174.899 \text{ cm}^2 \text{ Includes Bevel Area of } 58.300 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad ta = 38.7533 \text{ mm.} \\ \text{Wall Thickness per UG16(b),} & \quad tr_{16b} = 3.0000 \text{ mm.} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad trb_1 = 47.2220 \text{ mm.} \\ \text{Wall Thickness} & \quad tb_1 = \max(trb_1, tr_{16b}) = 47.2220 \text{ mm.} \\ \text{Wall Thickness} & \quad tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm.} \\ \text{Wall Thickness per table UG-45} & \quad tb_3 = 9.8312 \text{ mm.} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb_3, \max(tb_1, tb_2)] \\ &= \min[9.831, \max(47.222, 3.0)] \\ &= 9.8312 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(38.7533, 9.8312) \\ &= 38.7533 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073$ mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 48.0$, $t_r = 45.722$, $c = 1.5$ mm., $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.983$, Temp. Reduction = 1°C

Min Metal Temp. w/o impact per UCS-66, Curve B	16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	15 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 15 °C

Weld Size Calculations, Description: A1-Tube inlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000$ mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (225.6726 - 3.84 + 2 \cdot 158.5 \cdot 1.0 \cdot (1.0 \cdot 46.5 - 45.722)) \cdot 138) \\ &= 315623.69 \text{ Kg} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_4 - (W_i - C_{an} / 707)^2 \cdot fr_2) \cdot S_v \\ &= (223.1632 + 1. - 0.0 \cdot 1.0) \cdot 138 \\ &= 315432.75 \text{ Kg} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (223.1632 + 0.0 + 1. + (147.405)) \cdot 138 \\ &= 522854.72 \text{ Kg} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (223.1632 + 0.0 + 1. + 0.0 + (147.405)) \cdot 138 \end{aligned}$$



= 522854.72 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 810.5756 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 87791. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 326.0378) \cdot (160.0 - 1.5) \cdot 0.7 \cdot 138 \\ &= 1599144. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 810.5756 \cdot (46.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 609882. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (87791 + 1599144) = 1686936 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (87791 + 0 + 609882 + 0) = 697673 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (87791 + 609882 + 0) = 697673 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1686935 Kgf, must exceed W = 315623 Kgf or W1 = 315432 Kgf
Path 2-2 = 697672 Kgf, must exceed W = 315623 Kgf or W2 = 522854 Kgf
Path 3-3 = 697672 Kgf, must exceed W = 315623 Kgf or W3 = 522854 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 192 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 141.8250 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 389.8250 mm.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	191.00 bars
Shell Maximum Operating Pressure	Psox,max	191.00 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	98.0000 mm.
Shell Internal Corrosion Allowance	cas	1.5000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1250.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	85.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Bottom Dish

Channel Type:		Hemispherical Head
Channel Maximum Design Pressure	Ptd,max	191.00 bars
Channel Maximum Operating Pressure	Ptox,max	191.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	57.0000 mm.
Channel Internal Corrosion Allowance	cac	1.5000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1300.000 mm.
Channel Design Temperature	TEMPC	85.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	2893
Tube Wall Thickness	et	1.6500 mm.
Tube Outside Diameter	D	12.7000 mm.
Total Straight Tube Length	Lt	4960.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		4650.00 mm.
Design Temperature of the Tubes		85.00 °C
Tube Material		SA-210 A-1
Tube Material UNS Number		K02707
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	236.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	19.0500 mm.
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6500 mm.
Groove Weld Leg	ag	1.6500 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	536.453 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.
Area of Tube Layout	Ap	8856.5 cm ²
Perimeter of Tube Layout	Cp	3478.0 mm.

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 85.00 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	155.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm.
Tubesheet Outside Diameter	A	1446.000 mm.

Area of the Untubed Lanes	AL	0.0 cm ²
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Additional Data for Fixed/Floating Tubesheet Exchangers:

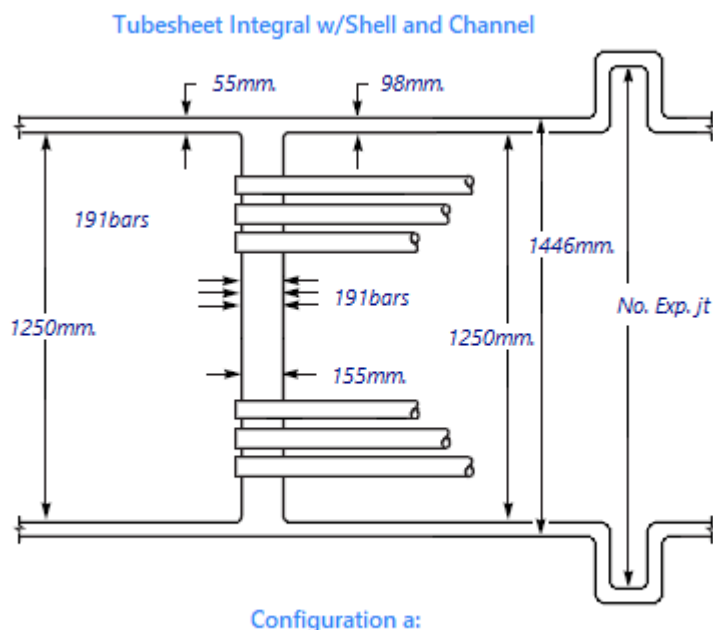
Unsupported Tube Span under consideration	l	500.000 mm.
Tube End condition corresponding to Span (l)	k	1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		Yes
Differential Design Pressure	Pdif	20.0000 bars
Run Multiple Load Cases		Yes

Mean Shell Metal Temp. along Shell len.	Tsm	21.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	30.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Tubesheet Integral with		Both
Tubesheet Extended as Flange		No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No
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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:



Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.0 °C	0.0000115174 /°C
Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	21.0 °C	0.20271E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	30.0 °C	0.0000116150 /°C
Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Metal Temp. along Len	30.0 °C	0.20219E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Differential pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (20.0 \cdot (12.7/2 - 0)) / (118.0 \cdot 1.0 + 0.4 \cdot 20.0)$
 $= 0.1069 + 0.0000 = 0.1069 \text{ mm.}$

Tube Required Thickness under External Pressure (Differential pressure) :

External Pressure Chart CS-2 at 85.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	12.70	4650.00	7.70	50.0000	0.0185675	122.73

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 243.2871 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3034	12.70	4650.00	41.85	50.0000	0.0006280	62.78

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 62.7848) / (3 \cdot 41.8524) = 20.0008 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3034 mm.
Allowable Internal Pressure at Corroded thickness	342.15 bars
Required Internal Design Pressure	20.00 bars
Allowable External Pressure at Corroded thickness	243.29 bars
Required External Design Pressure	20.00 bars
Required Thickness due to Shell Side pressure	0.3034 mm.

Note: The tubes were evaluated using the specified differential pressure.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)



Note: Shell/Channel Design Pressures modified by Input differential pressure:
Pdif = 20.000 Ps = 191.000 Pt = 171.000 bars

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:
 $= l_{tx} / h$ (modified for corrosion if present)
 $= 0/155.0$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:
 $= \max(dt - 2 \cdot tt (Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$
 $= \max(12.7 - 2 \cdot 1.65 (0.19904E+09/1.9904E+09) \cdot (117/137) \cdot (0); 12.7 - 2 \cdot 1.65)$
 $= 12.7000$ mm.

Equivalent outer tube limit circle diameter $[Do]$:
 $= 2 \cdot ro + dt$
 $= 2 \cdot 536.453 + 12.7$
 $= 1085.606$ mm.

Determine the basic ligament efficiency for shear $[\mu]$:
 $= (p - dt) / p$
 $= (19.05 - 12.7) / 19.05$
 $= 0.333$

Equivalent outer tube limit radius $[ao]$:
 $= Do / 2$
 $= 1085.606/2$
 $= 542.8030$ mm.

Effective tube pitch $[p^*]$:
 $= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$
 $= 19.05 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 1085.606 \cdot 19.05) / (\pi \cdot 1085.606^2)) }$
 $= 19.0500$ mm.

Effective ligament efficiency for bending $[\mu^*]$:
 $= (p^* - d^*) / p^*$
 $= (19.05 - 12.7) / 19.05$
 $= 0.3333$

Ratio $[ps]$:
 $= as / ao$
 $= 625.0/542.803$
 $= 1.1514$

Ratio $[pc]$:
 $= ac / ao$
 $= 650.0/542.803$
 $= 1.1975$

Compute parameter $[xt]$:
 $= 1 - Nt ((dt - 2 \cdot tt) / (2 \cdot ao))^2$
 $= 1 - 2893 ((12.7 - 2 \cdot 1.65) / (2 \cdot 542.803))^2$
 $= 0.7831$

Determine parameter $[xs]$:
 $= 1 - Nt (dt / (2 \cdot ao))^2$
 $= 1 - 2893 (12.7 / (2 \cdot 542.803))^2$
 $= 0.6041$

Determine the value $[h'g]$:



= max((hg - CATC); 0) (For pressure only cases)
 = max((6.0 - 0); 0)
 = 6.000 mm.

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

= $\pi \cdot ts (Ds + ts) Es / L$
 = $\pi \cdot 98.0 (1250.0 + 98.0) \cdot 1.9904E+09 / 4650.0$
 = 1811568.5000 Kg/mm.

Axial Tube Stiffness [Kt]:

= $\pi \cdot tt (Dt - tt) Et / L$
 = $\pi \cdot 1.65 (12.7 - 1.65) \cdot 1.9904E+09 / 4650.0$
 = 250.0259 Kg/mm.

Stiffness Factor [Ks,t]:

= $Ks / (Nt \cdot Kt)$
 = $1811568 / (2893 \cdot 250.026)$
 = 2.50450

Rigidity Ratio [J]:

= $1 / (1 + Ks/Kt)$
 = $1 / (1 + 1811568/0)$
 = 1.0000 (= 1 if No Exp. Jt.)

Shell coefficient [β_s]:

= $((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts)$
 = $((12(1 - 0.3^2))^{1/4}) / ((1250.0 + 98.0)98.0)$
 = 0.0050 1/mm.

Shell coefficient [ks]:

= $\beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$
 = $0.005 \cdot 0.19904E+09 \cdot 98.0^3 / (6(1 - 0.3^2))$
 = 1716044416.0000 bars·mm.²

Shell coefficient [λ_s]:

= $6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$
 = $6(1250.0 + 98.0) \cdot 0.17160E+1 / 155.0^3 \cdot (1 + 155.0 \cdot 0.005 + (155.0 \cdot 0.005)^2 / 2)$
 = 7736469.0000 bars

Shell coefficient [δ_s]:

= $Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$
 = $1250.0^2 / (4 \cdot 0.19904E+09 \cdot 98.0) \cdot (1 - 1250.0 \cdot 0.3 / (2(1250.0 + 98.0)))$
 = 0.17239E-01 mm./N./mm²

Channel coefficient [β_c]:

= $((12(1 - \nu_c^2))^{1/4}) / ((Dc + tc)tc)$
 = $((12(1 - 0.3^2))^{1/4}) / ((1300.0 + 57.0)57.0)$
 = 0.0065 1/mm.

Channel coefficient [kc]:

= $\beta_c \cdot Ec \cdot tc^3 / (6(1 - \nu_c^2))$
 = $0.007 \cdot 0.19904E+09 \cdot 57.0^3 / (6(1 - 0.3^2))$
 = 441271104.0000 bars·mm.²

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

CqDelta = CqTheta = CmTheta = CmDelta = 1

Channel coefficient [λ_c]:

= $6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$
 = $6(1300.0 + 57.0 / \cos(0)) \cdot 0.44127E+09 / 155.0^3 \cdot (1.0 + 155.0 \cdot 0.007 \cdot (1.0 + 1.0) / 2 + (155.0 \cdot 0.007)^2 / 2 \cdot 1.0)$
 = 2437418.8 bars

Channel coefficient [δ_c]:

= $Dc^2 / (4 \cdot Ec \cdot Tc) \cdot (1/2 - Dc \cdot \nu_c / (2(Dc + tc)))$
 = $1300.0^2 / (4 \cdot 0.19904E+09 \cdot 57.0) \cdot (1/2 - 1300.0 \cdot 0.3 / (2(1300.0 + 57.0)))$
 = 0.13268E-01 mm./N./mm²



Channel coefficient [zetac]:
zetac = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 8.136483$; $\mu^* = 0.333333$; $E^* = 64706924$ KPa.
 $E^*/E = 0.325091$; $\nu^* = 0.334167$

Note: As h/p (8.136) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:
$$= \left(\left(24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$
$$= \left(\left(24(1 - 0.334^2)2893 \cdot 0.19904E+09 \cdot 1.65 \right. \right. \\ \left. \left. (12.7 - 1.65)542.803^2 \right) / (64706924 \cdot 4650.0 \cdot 155.0^3) \right)^{1/4}$$
$$= 2.7698$$

Values from Table 4.18.3

$Z_d = 0.068773$; $Z_v = 0.120115$; $Z_m = 0.546768$
 $Z_a = 0.166668E+01$; $Z_w = 0.120115$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:
 $= A / Do$
 $= 1446.0/1085.606$
 $= 1.3320$

Compute coefficient [F]:
$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$
$$= (1 - 0.33) / 64706924 \cdot (7736469 + 2437418 + 0.19904E+09 \cdot \ln(1.33))$$
$$= 11.0560$$

Compute parameter [ϕ]:
$$= (1 + \nu^*)F$$
$$= (1 + 0.3342)11.056$$
$$= 14.7506$$

Compute parameter [Q1]:
$$= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$
$$= (1.1514 - 1 - 14.7506 \cdot 0.1201) / (1 + 14.7506 \cdot 0.5468)$$
$$= -0.178744$$

Compute Parameter [Qz1]:
$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot Xa^4$$
$$= (0.06877 + -0.17874 \cdot 0.12012) / 2 \cdot 2.76983^4$$
$$= 1.3921$$

Compute parameter [Qz2]:
$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot Xa^4$$
$$= (0.12012 + -0.17874 \cdot 0.54677) / 2 \cdot 2.76983^4$$
$$= 0.6587$$

Compute parameter [U]:
$$= (Z_w + (\psi_s - 1)Z_m)Xa^4 / (1 + \phi \cdot Z_m)$$
$$= (0.1201 + (1.1514 - 1)0.5468)2.76983^4 / (1 + 14.7506 \cdot 0.5468)$$
$$= 1.3175$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:
 $= 0$ (config a)

Compute Parameter [γ]:
 $= 0.000$ mm. (For Pressure only cases)

Calculate parameter [ω_s]:
$$= \psi_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$
$$= 1.1514 \cdot 0.17160E+1 \cdot 0.005 \cdot 0.017239 (1 + 155.0 \cdot 0.005)$$



$$= 30245.5859 \text{ mm.}^2$$

Calculate parameter $[\omega*s]$:

$$\begin{aligned} &= ao^2(\rho s^2 - 1) \cdot (\rho s - 1) / 4 - \omega s \\ &= 542.803^2(1.151^2 - 1) \cdot (1.151 - 1) / 4 - 30245.586 \\ &= -26611.6328 \text{ mm.}^2 \end{aligned}$$

Parameter $[\omega c]$:

$$\begin{aligned} &= \rho c \cdot kc[\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\ &\quad \zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)) \\ &= 1.197 \cdot 0.44127E+09[0.013268 (1.0 + 0.007 \cdot 1.0 \cdot 155.0) - \\ &\quad 0 (1.0/155.0 + 1.0 \cdot 0.007/2)) \\ &= 9225.6553 \text{ mm.}^2 \end{aligned}$$

Parameter $[\omega*c]$:

$$\begin{aligned} &= ao^2[(\rho c^2 + 1) \cdot (\rho c - 1) / 4 - (\rho s - 1) / 2] - \omega c \\ &= 542.80298^2[(1.19749^2 + 1) \cdot (1.19749 - 1) / 4 - (1.15143 - 1) / 2] - 9225.65527 \\ &= 3872.3215 \text{ mm.}^2 \end{aligned}$$

Pressure $[P*S]$:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure $[P*C]$:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure $[P's]$:

$$\begin{aligned} &= Ps\{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - \\ &\quad [(\rho s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(Dj)^2 - (Ds)^2 / Do^2] \} \\ &= 191.0 \{ 0.604 + 2(1 - 0.604) 0.3 + [2/2.505 \cdot \\ &\quad (1250.0/1085.606)^2] 0.3 - [(1.151^2 - 1) / (1.0 \cdot 2.505)] - \\ &\quad [(1 - 1.0) / (2 \cdot 1.0 \cdot 2.505)] [(0^2 - (1250.0)^2 / 1085.606^2)] \} \\ &= 196.5707 \text{ bars} \end{aligned}$$

Pressure $[P't]$:

$$\begin{aligned} &= [xt + 2(1 - xt) vt + 1 / (J \cdot Kst)] Pt \\ &= [0.783 + 2(1 - 0.783) 0.3 + 1 / (1. \cdot 2.505)] 171.0 \\ &= 224.4411 \text{ bars} \end{aligned}$$

Pressure $[P_Y]$:

$$\begin{aligned} &= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2) \\ &= 2893 \cdot 250.026 \cdot 0 / (3.142 \cdot 542.803^2) \\ &= 0.0000 \text{ bars} \end{aligned}$$

Pressure $[P_w]$:

$$\begin{aligned} &= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2) \\ &= -0 \cdot 1.317 \cdot 0 / (2 \cdot \pi \cdot 542.803^2) \\ &= -0.0000 \text{ bars} \end{aligned}$$

Calculate the pressure $[Prim]$:

$$\begin{aligned} &= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt) \\ &= -(1.317/542.803^2) \cdot (-26611.633 \cdot 191.0 - 3872.322 \cdot 171.0) \\ &= 25.6892 \text{ bars} \end{aligned}$$

Calculate the pressure $[P_w]$:

$$\begin{aligned} &= (U/ao^2) \cdot (\omega s \cdot P*s - \omega c \cdot P*c) \\ &= (1.317/542.803^2) \cdot (30245.5859 \cdot 0 - 9225.6553 \cdot 0) \\ &= 0.0000 \text{ bars} \end{aligned}$$

Determine the effective pressure $[Pe]$:

$$\begin{aligned} &= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (\rho s - 1) Qz2)) \cdot \\ &\quad (P's - P't + P_Y + P_w + P_w + Prim) \\ &= 0.1000E+01 \cdot 2.505 / (1 + 1.0 \cdot 2.505 (1.392 + (1.151 - \\ &\quad 1) 0.659)) \cdot (196.571 - 224.441 + 0 + 0 + 0 + 25.689) \\ &= -1.1534 \text{ bars} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor $[Q2]$:



$$= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega s \cdot P * s - \omega c \cdot P * c)) + W * \cdot y_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-26611.633 \cdot 191.0 - 3872.322 \cdot 171.0) - (30245.586 \cdot 0 - 9225.655 \cdot 0)) \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 14.7506 \cdot 0.54677)$$

$$= -6462.6426 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2)$$

$$= -0.179 + 2 \cdot -6462.643 / (-1.153 \cdot 542.803^2)$$

$$= 3.55112$$

Fm Value from Table 4.18.3 = 1.799438

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 1.7994 / 0.3333) \cdot (2 \cdot 542.803 / (155.0 - 6.0))^2 \cdot -1.15$$

$$= -49.5807 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138.0 = 207.0 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.0076 / 0.3333) \cdot (2 \cdot 542.803 / (10.005 - 6.0))^2 \cdot -0.82$$

$$= -206.9856 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 10.0046 + 0 + 0$$

$$= 10.0046 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [Tlimit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.333 \cdot 155.0 / 542.803) \cdot 110.398$$

$$= 21.0165 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm.}$$

$$\text{Tubesheet Shear Stress} = 10.7677 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(HreqB; HreqS)$$

$$= \max(10.0046 ; 3.81)$$

$$= 10.0046 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$Ftmin = -10.4104, Ftmax = 14.5769$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - Pe \cdot Ftmin) / (X_t - X_s)$$

$$= ((191.0 \cdot 0.6041 - 171.0 \cdot 0.7831) - (-1.153) \cdot -10.41) / (0.7831 - 0.6041)$$

$$= -17.0594 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - Pe \cdot Ftmax) / (X_t - X_s)$$

$$= ((191.0 \cdot 0.6041 - 171.0 \cdot 0.7831) - (-1.153) \cdot 14.577) / (0.7831 - 0.6041)$$

$$= -0.9603 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(| \sigma_{t,1} | ; | \sigma_{t,2} |) = 17.059 \text{ N./mm}^2$$



Allowable Tube Stress [σ_t, A]:
= Sot = 117.9979 N./mm²

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
= max((3.25 - 0.25(Z_d + Q₃ • Z_w)X_a⁴); 1.25)
= max((3.25 - 0.25(0.069 + 3.551 • 0.12)2.77⁴); 1.25)
= 1.2500 (Should be ≤ 2)

Determine the factor [rt]:
= $(\sqrt{ \frac{dt^2 + (dt - 2 \cdot tt)^2 }{4} })$
= $(\sqrt{ \frac{12.7^2 + (12.7 - 2 \cdot 1.65)^2 }{4} })$
= 3.9501 mm.

Determine the factor [Ct]:
= $\sqrt{ \frac{2 \cdot \pi^2 \cdot Et}{Syt} }$
= $\sqrt{ \frac{2 \cdot \pi^2 \cdot 0.19904E+09}{235} }$
= 129.0317

Determine the factor [Ft]:
= k • L/r
= 1.0 • 500.0/3.95
= 126.5797

Buckling Allowable Stress [Stb]:
= (Sy,t/Fs) • (1 - Ft/(2 • Ct))
= (235/1.25) • (1 - 126.58/(2 • 129.032))
= 96.193 N./mm² (Never greater than Sot)

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
= $\sigma_{t,max} \cdot \text{Tube Area}$
= 17.06 • 0.5728
= 99.64 Kgf

Tube Metal Area [At]:
= $\pi(d_o - t)t$
= $\pi(12.7 - 1.65)1.65$
= 0.573 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
= $\pi \cdot t(d_o - t)S$
= $\pi \cdot 1.65 (12.7 - 1.65)117.998$
= 689.194 Kgf

Groove Weld Strength [Fg]:
= $0.85 \cdot \pi \cdot a_g(d_o + 0.67 \cdot a_g)S_w$ (but not > Ft)
= $0.85 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65)118.0$
= 689.1942 Kgf

Weld Strength Factor [fw]:
= S / S_w where S_w = min(S and S_t)
= 117.998/117.998
= 1.000

Design Strength Ratio [fd]:
= 1 for full strength welds

Fillet Weld Strength [Ff]:
= $0.55 \cdot \pi \cdot a_f(d_o + 0.67 \cdot a_f)S_w$ (but not > Ft)
= $0.55 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65)118.0$
= 473.5809 Kgf

Minimum Length of Weld [ar]:
= $2(\sqrt{ (0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t)fw \cdot fd }) - 0.75 \cdot d_o$



$$= 2 \cdot \sqrt{(0.75 \cdot 12.7)^2 + 1.07 \cdot 1.65 \cdot (12.7 - 1.65) \cdot 1.0 \cdot 1.0} - 0.75 \cdot 12.7$$

$$= 1.949 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 0.9900 mm.
 Minimum Required Groove Weld Leg [agr]: 0.9900 mm.

Tube-Tubesheet Jt allowable, 689.19 is >= tube strength 689.19 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= S_s \cdot E_{sw} = 138.0 \cdot 1.0 = 138.0 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 542.803^2 / ((1250.0 + 98.0) 98.0) \cdot [-1.15 + (1.151^2 - 1)]$
 $+ (191.0 - 171.0) / (625.0^2 \cdot 171.0 / ((1250.0 + 98.0) 98.0))$
 $= 51.7629 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 98.0 mm., the minimum Shell length adjacent to the tubesheet should be 630.000 mm.

Shell Membrane Stress due to Joint Interaction [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 542.803^2 / ((1250.0 + 98.0) 98.0) [-1.15 + (1.151^2 - 1)]$
 $+ (191.0 - 171.0) / (625.0^2 \cdot 171.0 / ((1250.0 + 98.0) 98.0))$
 $= 51.7629 \text{ N./mm}^2$

Shell Bending Stress due to Joint Interaction [σ_s, b]:
 $= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E \cdot (a_o/h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2/a_o^2 \cdot Z_m \cdot Q_2]] \}$
 $= 6 \cdot 0.17160 \text{E}+1 / 98.0^2 \{ 0.005 [0.017239 \cdot 191.0 + 625.0^2 \cdot 0 / (1.9506 \text{E}+11)] + 6(1 - 0.33^2) / (64706924) (542.8 / 155.0)^3 (1 + 155.0 \cdot 0.01 / 2) [-1.2 (0.12 + 0.547 \cdot -0.179) + 2 / 542.8^2 \cdot 0.547 \cdot -6462.643] \}$
 $= 51.4240 \text{ N./mm}^2$

Shell Stress Summation vs. Allowable
 $:: |\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot S_s$
 $:: |51.76| + |51.42| \leq 207.0 \text{ N./mm}^2$
 $:: 103.19, \text{ must be } \leq 207.0 \text{ N./mm}^2$

Channel Membrane Stress due to Joint Interaction [σ_c, m]:
 $= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$
 $= 650.0^2 \cdot 171.0 / ((1300.0 + 57.0) 57.0 \cdot \cos(0))$
 $= 93.410 \text{ N./mm}^2$

Channel Bending Stress due to Joint Interaction [σ_c, b]:
 $= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^2) / E \cdot (a_o/h)^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta)) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2/a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 0.44127 \text{E}+09 / 57.0^2 \cdot \{ (0.013268 \cdot 1.0 \cdot 0.007 - 0 \cdot 1.0) \cdot 171.0 - 6((1 - 0.334^2) / 64706924) \cdot (542.803 / 155.0)^3 \cdot (1.0 + 155.0 \cdot 0.007 / 2 \cdot 1.0) \cdot [-1.2 (0.12012 + 0.54677 \cdot -0.179) + 2 / 542.803^2 \cdot 0.54677 \cdot -6462.643] \}$
 $= 224.1423 \text{ N./mm}^2$

Channel Stress Summation vs. Allowable
 $:: |\sigma_{c,m}| + |\sigma_{c,b}| \leq 1.5 \cdot S_c$
 $:: |93.41| + |224.14| \leq 207.0 \text{ N./mm}^2$
 $:: 317.55 \text{ must be } \leq 207.0 \text{ N./mm}^2 \text{ (See Elast-Plast iteration below)}$

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	49.6 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	0.6 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	17.1 <=	118.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-17.1 <=	-96.2 N./mm ²	Pass
Maximum Force on any one Tube	99.6 <=	689.2 Kg _f	Pass
Axial Membrane Stress in Shell	51.8 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	103.2 <=	207.0 N./mm ²	Pass
Channel Stress (jt. inter.)	317.6 <=	207.0 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:
 = min(Sys, SPS,s/2)
 = min(242, 484/2) = 242.01 N./mm²

Compute Allowable [S*c]:
 = min(Syc, SPS,c/2)
 = min(242, 484/2) = 242.01 N./mm²

Compute Factor [facts]:
 = min([1.4 - 0.4* abs(σs,b) / Sstars], 1)
 = min([1.4 - 0.4* abs(51.42)/242.01], 1) = 1.0

Compute Factor [factc]:
 = min([1.4 - 0.4* abs(σc,b) / Sstarc], 1)
 = min([1.4 - 0.4* abs(224.14)/242.01], 1) = 1.0

Note: Since facts & factc are both >= 1, the design Passes.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Shell Stress (jt. inter.)	103.2 <=	207.0 N./mm ²	Pass
Channel Stress (jt. inter.)	317.6 <=	484.0 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	10.005	155.000	Ok
Tube-Tubesheet Fillet Weld Leg :	0.990	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	0.990	1.650	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D3uc	10.005	...	50	207	1	110	Ps+Pt-Th D3	Ok
O3uc	6.884	...	84	459	9	110	Ps+Pt+Th O3	Ok
O4uc	8.390	...	39	459	10	110	Ps+Pt+Th O4	Ok
D3c	10.864	...	51	207	...	110	Ps+Pt-Th-c D3	Ok
O3c	10.158	...	87	459	9	110	Ps+Pt+Th-c O3	Ok
O4c	11.357	...	39	459	10	110	Ps+Pt+Th-c O4	Ok



Max: 11.3568 ... mm. 0.247 0.088 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D3uc	52	137	...	...	:	...	...	...	...	Ok
O3uc	56	484	...	...	:	...	...	...	...	Ok
O4uc	4	484	...	...	:	...	...	...	...	Ok
D3c	53	137	...	...	:	...	...	...	...	Ok
O3c	57	484	...	...	:	...	...	...	...	Ok
O4c	4	484	...	...	:	...	...	...	...	Ok
Max RATIO	0.383				:	...				...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D3uc	17	117	-17	-96	100	689	103	206	318	484	Ok
O3uc	22	235	-22	-60	130	1378	131	484	298	484	Ok
O4uc	15	235	-15	-60	89	1378	28	484	19	484	Ok
D3c	17	117	-17	-96	101	689	104	206	327	484	Ok
O3c	22	235	-22	-60	130	1378	132	484	307	484	Ok
O4c	15	235	-15	-60	89	1378	28	484	20	484	Ok
Max RATIO	0.147		0.371		0.147		0.502		0.675		

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	11.357	155.000	Ok
Tube thickness :	0.303	1.650	Ok
Tube-tubesheet fillet weld leg :	0.990	1.650	Ok
Tube-tubesheet groove weld leg :	0.990	1.650	Ok

Min Shell length of thk, (98.000) adj. to tubesheet: 630.000 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.486 mm.



Due to Pressure Shell Compresses by : -0.472 mm.
 Due to Pressure + Thermal Shell Extends by : +.486 mm.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	144.991	1.000	134.913	1.000
Tubesheet Shear Stress	325.477	1.000	366.180	1.000
Tube Tensile Stress	397.005	1.000	672.856	1.000
Tube Compressive Stress	202.275	1.000	630.608	1.000
Tube-Tubesheet Joint load	397.004	1.000	672.849	1.000
Shell Stress (Axial, Junction)	144.991	0.701	134.913	0.364
Tubesheet-Channel Junction	144.991	0.615	134.913	0.035
Tube Pressure Stress	342.152	1.000	243.286	1.000
Minimum MAWP	144.991		134.913	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	144.991	1.000	134.913	1.000
Tubesheet Shear Stress	327.039	1.000	368.923	1.000
Tube Tensile Stress	414.558	1.000	712.917	1.000
Tube Compressive Stress	219.348	1.000	654.877	1.000
Tube-Tubesheet Joint load	414.558	1.000	712.917	1.000
Shell Stress (Axial, Junction)	144.991	0.673	134.913	0.331
Tubesheet-Channel Junction	144.991	0.554	134.913	0.030
Tube Pressure Stress	342.152	1.000	243.286	1.000
Minimum MAPnc	144.991		134.913	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure UnCorroded	Due to Pressure Corroded	Due to Thermal Exp. UnCorroded	Due to Thermal Exp. Corroded	Computed UnCorroded	Delta Ms Corroded
1	0.0924	0.0924	...	...	...	...
2	-0.4717	-0.4720	...	...	...	...
3	-0.3890	-0.3892	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	-0.3890	-0.3892	0.4860	0.4864	...	...
8	...	...	0.4860	0.4864	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(155.0/4, \min(155.0, 98.0))$$

$$= 98.000 \text{ mm.}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(191.0/134.91, 1)$$

$$= 1.000$$

Thickness Ratio = 1.0, Temperature Reduction per Fig. UCS 66.1 = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 31 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit 31 °C



Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(155.0/4, min(155.0, 57.0))
= 57.000 mm.

MDMT reduction ratio per UCS 66 (b)(1)(b):
= min(pt/Tubeside MAPnc, 1)
= min(191.0/144.99, 1)
= 1.000

Thickness Ratio = 1.0, Temperature Reduction per Fig. UCS 66.1 = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	20 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	20 °C

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